SIKSHA 'O' ANUSANDHAN
DEEMED TO BE UNIVERSITY

DevOps Summer Training Program 2025

Submitted by-

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Admission Batch: 2022

Session: 2025-2026

Serial No.: 47

Department of Computer Science & Engineering Faculty of Engineering & Technology (ITER)

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Assignment 1
TOPIC : Basic Linux Commands
Date: 24/06/25

Navigation & Directory Management:

Question 01:

What command will take you to your home directory?

Command:

cd ~

Question 02:

How do you move to the /etc directory?

Command:

cd /etc

Question 03:

List all files (including hidden ones) in the current directory.

Command:

ls -a

Question 04:

Create a new directory named my_folder to your home directory.




```
ubuntu@ip-172-31-31-201:~$ mkdir myfolder
ubuntu@ip-172-31-31-201:~$ ls
myfolder
ubuntu@ip-172-31-31-201:~$
```

Question 05:

Rename the directory my_folder to my_documents.

```
ubuntu@ip-172-31-31-201:~$ mv myfolder my_documents
ubuntu@ip-172-31-31-201:~$ ls
my_documents
ubuntu@ip-172-31-31-201:~$
```

Question 06:

Delete & directory called test folder (assume it's empty).

```
ubuntu@ip-172-31-31-201:~$ ls
my_documents  test
ubuntu@ip-172-31-31-201:~$ rmdir test
ubuntu@ip-172-31-31-201:~$ ls
my_documents
ubuntu@ip-172-31-31-201:~$
```

1.

File Operations:

Question 07:

Create a new empty file named test.txt

```
ubuntu@ip-172-31-31-201:~$ touch test.txt
ubuntu@ip-172-31-31-201:~$ ls
my_documents  test.txt
ubuntu@ip-172-31-31-201:~$
```

Question 08:

Copy test.txt to a file called test_backup.txt.

```
ubuntu@ip-172-31-31-201:~$ cp test.txt test_backup.txt
ubuntu@ip-172-31-31-201:~$ cat test_backup.txt
hi this is test.txt
ubuntu@ip-172-31-31-201:~$ cat test.txt
hi this is test.txt
ubuntu@ip-172-31-31-201:~$
```

Question 09:

Move test_backup.txt into a folder named backups.

```
ubuntu@ip-172-31-31-201:~$ mv test_backup.txt backups/
ubuntu@ip-172-31-31-201:~$ cd backups
ubuntu@ip-172-31-31-201:~/backups$ ls
test_backup.txt
ubuntu@ip-172-31-31-201:~/backups$
```

Question 10:

Remove the file test.txt.

```
ubuntu@ip-172-31-31-201:~$ rm test.txt
ubuntu@ip-172-31-31-201:~$ ls
backups  my_documents
ubuntu@ip-172-31-31-201:~$
```

Question 11:

Use a command to view the contents of a file named notes.txt.

```
ubuntu@ip-172-31-31-201:~$ cat notes.txt
hi, this is shruti
this is my file notes.txt
ubuntu@ip-172-31-31-201:~$
```

Question 12:

Count the number of lines in the file notes.txt.

```
ubuntu@ip-172-31-31-201:~$ wc -l notes.txt
2 notes.txt
ubuntu@ip-172-31-31-201:~$
```

Permissions

Question 13:

Change the permissions of a file script.sh to make it executable.

```
ubuntu@ip-172-31-31-201:~$ chmod +x script.sh
ubuntu@ip-172-31-31-201:~$
```

Question 14:

Display the permissions of all files in the current directory.

```
ubuntu@ip-172-31-31-201:~$ ls -l
total 12
drwxrwxr-x 2 ubuntu ubuntu 4096 Aug  5 18:05 backups
drwxrwxr-x 2 ubuntu ubuntu 4096 Aug  5 18:01 my_documents
-rw-rw-r-- 1 ubuntu ubuntu   45 Aug  5 18:08 notes.txt
-rwxrwxr-x 1 ubuntu ubuntu    0 Aug  5 18:10 script.sh
ubuntu@ip-172-31-31-201:~$
```

Question 15:

Change ownership of the file report.txt to User "admin"

sudo chown admin report.txt

Compression & Archiving:

Question 16:

Create a tar archive of your Documents folder and name it documents_backup.tar.

```
ubuntu@ip-172-31-31-201:~$ tar -cvf documents_backup.tar Documents
Documents/
ubuntu@ip-172-31-31-201:~$ ls
Documents  backups  documents_backup.tar  my_documents  notes.txt  script.sh
ubuntu@ip-172-31-31-201:~$
```

Question 17:

Extract a tar file named code_archive.tar into the current directory.

```
ubuntu@ip-172-31-31-201:~/backups$ tar -xvf code_archive.tar
code_archive/
ubuntu@ip-172-31-31-201:~/backups$
```

Question 18:

Compress a file log.txt using gzip.

```
ubuntu@ip-172-31-31-201:~/backups$ gzip log.txt
```

Question 19:

Decompress a file log.txt.gz

```
ubuntu@ip-172-31-31-201:~/backups$ gunzip log.txt.gz
```

Question 20:

List the contents of an archive file named backup.tar.gz without extracting it.

```
ubuntu@ip-172-31-31-201:~/backups$ tar -tzf backup.tar.gz
```

Date-Based Task:

Question 21:

Create a backup of your home directory named as today's date (eg., 2025-06-24.tar).

tar -cvf ~/$(date +%F).tar

Question 22:

Copy that archive to the /root director

sudo cp ~/$(date +%F).tar /root/

Question 23:

Extract the archive inside /root.

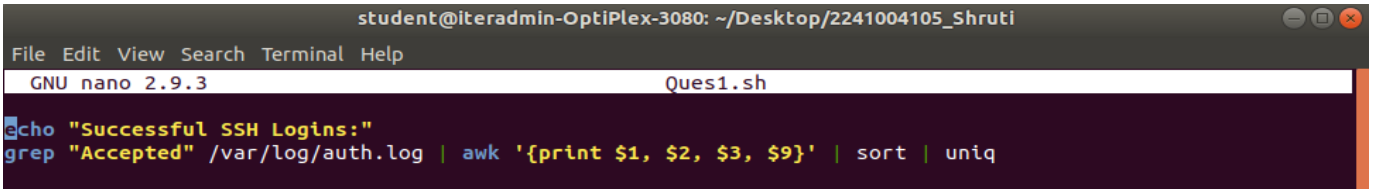
sudo tar -xvf /root/$(date +%F).tar -C /root/

ASSIGNMENT 2
TOPIC: BASH SCRIPTING

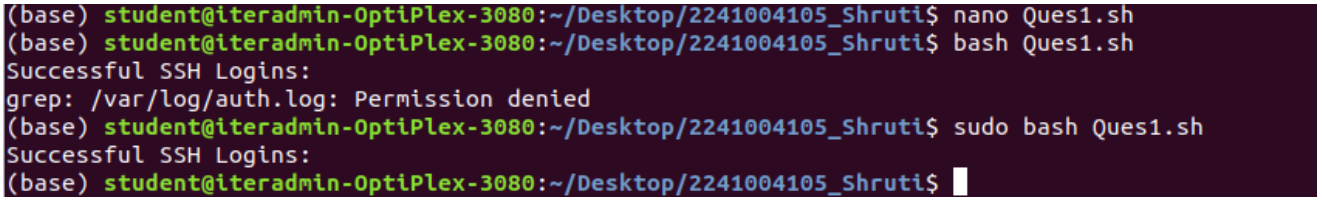
Name: Shruti Das
Regd. No: 2241004105
Date : 01/07/2025

Question 01:
User Login Tracker
Write a script to read the /var/log/auth.log file and extract the list of successful SSH login attempts using grep, awk, or cut. Display the unique usernames and their login timestamps.

Bash script:

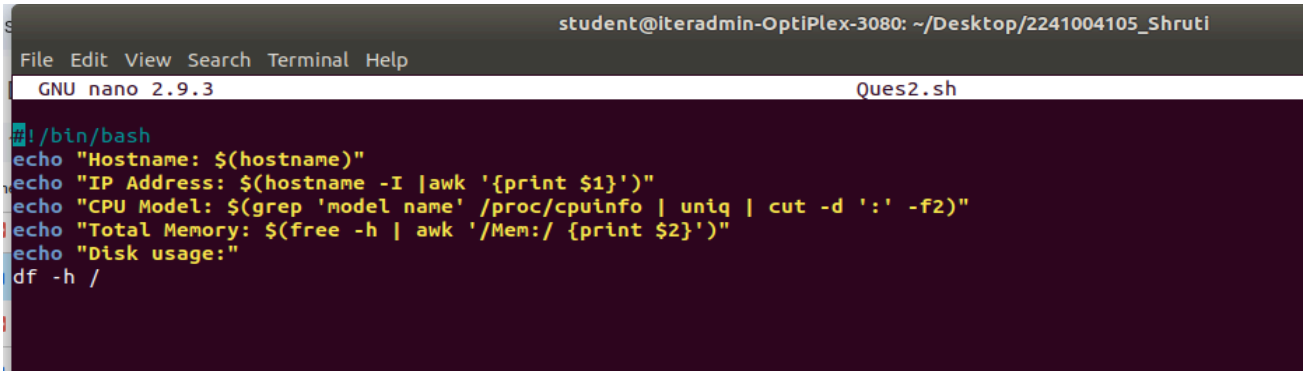


Output:

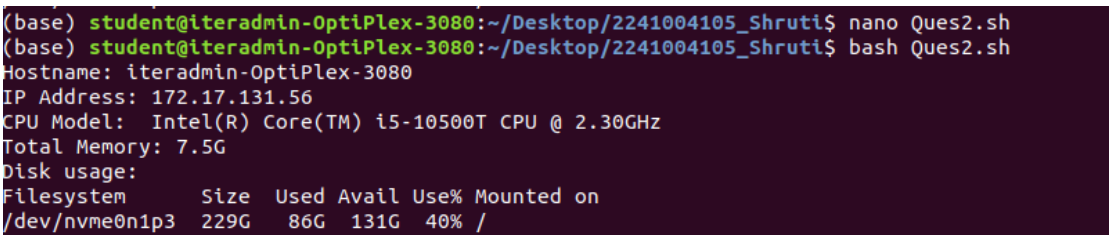


Question 02:
System Inventory Report
Create a script that gathers and displays system information: hostname, IP address, CPU model, total memory, and disk usage. Use commands like uname, hostname, ifconfig, free, and df.

Bash Script:



Output:

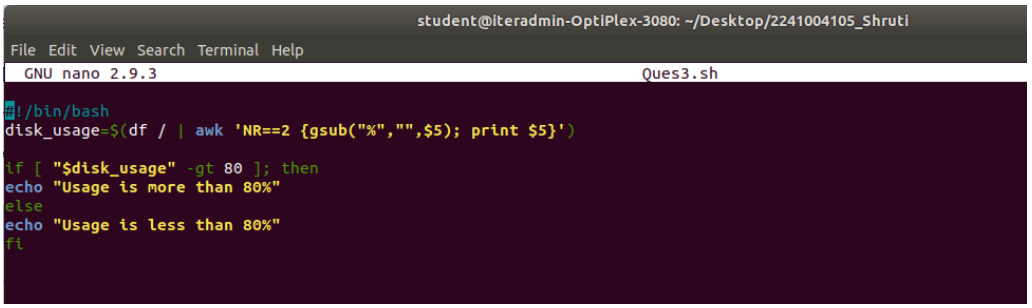


Question 03:

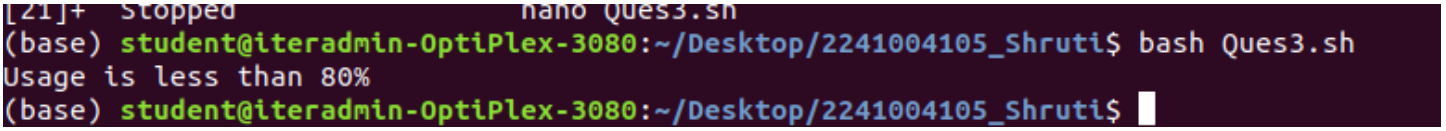
Check Disk Usage & Alert **6.**

Write a script that checks the disk usage of root (/) partition. If the usage exceeds 80%, print an alert. Use df, awk, and if conditionals.

Bash Script:



Output:

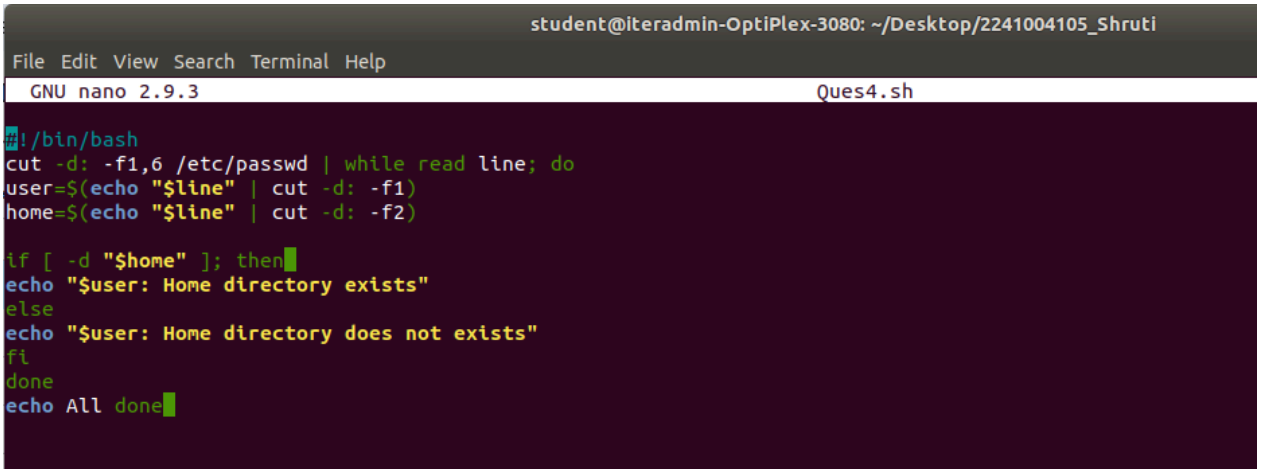


Question 04:

User Home Directory Checker

Loop through all users in /etc/passwd, extract their usernames and home directories using cut, and print whether the directory exists or not using an if check.

Bash Script:



Output:

```
[22]+  Stopped                 nano Ques4.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti$ bash Ques4.sh
root: Home directory exists
daemon: Home directory exists
bin: Home directory exists
sys: Home directory exists
sync: Home directory exists
games: Home directory exists
man: Home directory exists
lp: Home directory does not exists
mail: Home directory exists
news: Home directory does not exists
uucp: Home directory does not exists
proxy: Home directory exists
www-data: Home directory exists
backup: Home directory exists
list: Home directory does not exists
irc: Home directory does not exists
gnats: Home directory does not exists
nobody: Home directory does not exists
systemd-network: Home directory exists
systemd-resolve: Home directory exists
syslog: Home directory does not exists
messagebus: Home directory does not exists
_apt: Home directory does not exists
uuidd: Home directory exists
dnsmasq: Home directory exists
avahi-autoipd: Home directory exists
usbmux: Home directory does not exists
whoopsie: Home directory does not exists
kernoops: Home directory exists
rtkit: Home directory exists
avahi: Home directory exists
cups-pk-helper: Home directory does not exists
saned: Home directory does not exists
speech-dispatcher: Home directory does not exists
colord: Home directory exists
pulse: Home directory does not exists
hplip: Home directory does not exists
geoclue: Home directory exists
gnome-initial-setup: Home directory does not exists
gdm: Home directory exists
iteradmin: Home directory exists
```

Question 05:
Parse and Filter Logs
Using grep and awk, write a script to parse a given log file and list all lines containing the word "ERROR" along with the line number.

Bash Script:

```
student@iteradmin-OptiPlex-3080: ~/Desktop/2241004105_Shruti
File  Edit  View  Search  Terminal  Help
 GNU nano 2.9.3                                    Ques5.sh

#!/bin/bash
logfile=$1
grep -n "error" "$logfile"
```

Output:

```
[16]+  Stopped                 nano Ques5.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti$ bash Ques5.sh /var/log/bootstrap.log
305:Selecting previously unselected package libgpg-error0:amd64.
306:Preparing to unpack .../libgpg-error0_1.27-6_amd64.deb ...
307:Unpacking libgpg-error0:amd64 (1.27-6) ...
576:Setting up libgpg-error0:amd64 (1.27-6) ...
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti$ nano Ques5.sh
```

Question 06:
Service Status Checker
Create a script that takes a list of service names and checks whether each service is active or inactive using systemctl or service. Use loops and if statements.

Bash Script:

```
student@iteradmin-OptiPlex-3080: ~/Desktop/2241004105_Shruti
File  Edit  View  Search  Terminal  Help
  GNU nano 2.9.3                                  Ques6.sh

#!/bin/bash
read -p "Enter Services separated by space" -a services

for service in "${services[@]}"; do
systemctl is-active --quiet "$service"

if [ $? -eq 0 ]; then
echo "$service is running"
else
echo "$service is not running"
fi
done
```

Output:

```
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti$ bash Ques6.sh
Enter Services separated by spacecron fake
cron is running
fake is not running
```

Question 07:

CSV Data Cleaner

Given a CSV file with extra spaces and inconsistent delimiters, use awk, sed, and cut to clean it and print only selected columns neatly.

Bash Script:

```
student@iteradmin-OptiPlex-3080: ~/Desktop/2241004105_Shruti
File  Edit  View  Search  Terminal  Help
  GNU nano 2.9.3                                  Ques7.sh

#!/bin/bash
sed 's/ *, */,/g' new_data.csv | awk -F, '{print $1, $2, $3 }'
```

Output:

```
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti$ nano Ques7.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti$ bash Ques7.sh
name age department
shruti 21 aeroscience
sweta 32 physics
uskan 28 maths
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti$
```

CSV_file:

```
student@iteradmin-OptiPlex-3080: ~/Desktop/2241004105_Shruti
File  Edit  View  Search  Terminal  Help
  GNU nano 2.9.3                                  new_data.csv

name ,   age, department
shruti, 21 , aeroscience
sweta, 32  ,   physics
uskan, 28,  maths
```

Question 08:
Password Strength Validator
Accept a string as input and use a combination of regex (grep or [[... =~]]) to validate that it's at least 8 characters, contains one number, and one special character.

Bash Script:

```
student@iteradmin-OptiPlex-3080: ~/Desktop/2241004105_Shruti
File  Edit  View  Search  Terminal  Help
  GNU nano 2.9.3                                  Ques8.sh

#!/bin/bash
read -sp "Enter Password:" password
echo

if [[ ${#password} -ge 8 && $password =~ [0-9] && $password =~ [^a-zA-Z0-9] ]]; then
echo "Strong password"
else
echo "Weak password"
echo "Password must be 8 characters, with a number and a special character"
fi
```

Output:

```
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti$ nano Ques8.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti$ bash Ques8.sh
Enter Password:
Strong password
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti$ bash Ques8.sh
Enter Password:
Weak password
Password must be 8 characters, with a number and a special character
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti$
```

Question 09:
Process Monitor
Create a script that accepts a process name as input, checks if it is running using ps and grep, and displays its PID and memory usage.

Bash Script:

```
student@iteradmin-OptiPlex-3080: ~/Desktop/2241004105_Shruti
File  Edit  View  Search  Terminal  Help
  GNU nano 2.9.3                                  Ques9.sh

#!/bin/bash
read -p "Enter a process name:" Proccess_id

ps aux | grep "$Process_id" | grep -v grep | awk '{print "PID:", $2. "Memory:" $4"%"}'
```

Output:

```
student@iteradmin-OptiPlex-3080: ~/Desktop/2241004105_Shruti
File  Edit  View  Search  Terminal  Help
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti$ nano Ques9.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti$ ps
  PID TTY          TIME CMD
 8150 pts/1    00:00:00 bash
 8166 pts/1    00:00:00 nano
 8823 pts/1    00:00:00 nano
 8880 pts/1    00:00:00 nano
 8964 pts/1    00:00:00 nano
 9277 pts/1    00:00:00 nano
 9420 pts/1    00:00:00 ps
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti$ bash Ques9.sh 8150
Enter a process name:8150
PID: PIDMemory:%MEM%
PID: 1Memory:0.1%
PID: 2Memory:0.0%
PID: 3Memory:0.0%
PID: 4Memory:0.0%
PID: 6Memory:0.0%
PID: 8Memory:0.0%
PID: 9Memory:0.0%
PID: 10Memory:0.0%
PID: 11Memory:0.0%
PID: 12Memory:0.0%
PID: 14Memory:0.0%
```

Question 10:
Scheduled Cleanup
Write a script that deletes .log files older than 7 days from a given directory. Use find, conditionals, and optionally log what files were removed.

Bash Script:

```
student@iteradmin-OptiPlex-3080: ~/Desktop/2241004105_Shruti
File  Edit  View  Search  Terminal  Help
  GNU nano 2.9.3                          Ques10.sh

#!/bin/bash
read -p "Enter Directory to clean:" dir_name
if [[ ! -d "$dir_name" ]]; then
echo "Directory not found"
exit 1
fi

echo "Deleted log files:" > files_deleted.txt
find "$dir_name" -name "*.log" -type f -mtime +7 -print -delete >>files_deleted.txt
echo "Cleanup Complete"
```

Output:

```
PID: 9525Memory:0.0%
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti$ nano Ques10.sh
Use "fg" to return to nano.

[7]+  Stopped                 nano Ques10.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti$ bash Ques10.sh
Enter Directory to clean:/home/student/Documents/Ravi/bashscripts
Cleanup Complete
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti$
```

ASSIGNMENT-3
TOPIC: REGEX

Name: Shruti Das
Regd. No. : 2241004105
Date: 02/07/2025

Question 01:
From datafile find the name age and department of all employees whose name starts with ‘s’ and ends with ‘a’.
Example:- awk '/^J/{ print $0 }' datafile.

CSV FILE:

```
student@iteradmin-OptiPlex-3080: ~/Desktop/2241004105_Shruti/Assignment 2
File  Edit  View  Search  Terminal  Help
  GNU nano 2.9.3                                  Data.csv

Name,Age,Department
Sita,25,HR
Sweta,19,IT
Shruti,21,DevOps
John,34,Development
Ratan,26,Finance
Mira,18,Arts
```

Bash Script:

```
student@iteradmin-OptiPlex-3080: ~/Desktop/2241004105_Shruti/Assignment 2
File  Edit  View  Search  Terminal  Help
  GNU nano 2.9.3                                  Ques1.sh

#!/bin/bash
awk -F, 'tolower($1) ~ /^s.*a$/' Data.csv
```

Output:

```
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ nano Data.csv
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ nano Ques1.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ bash Ques1.sh
Sita,25,HR
Sweta,19,IT
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$
```

Question 02:
Print all records where the record ends with n.

Bash Script:

```
student@iteradmin-OptiPlex-3080: ~/Desktop/2241004105_Shruti/Assignment 2
File  Edit  View  Search  Terminal  Help
  GNU nano 2.9.3                                  Ques2.sh

#!/bin/bash

awk -F, 'tolower($1) ~ /n$/' Data.csv
```

Output:

```
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ nano Ques2.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ bash Ques2.sh
John,34,Development
Ratan,26,Finance
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$
```

Question 03:
Print all records that start with J and ends with n.

Bash Script:

```
                                    student@iteradmin-OptiPlex-3080: ~/Desktop/2241004105_Shruti/Assignment 2
File  Edit  View  Search  Terminal  Help
  GNU nano 2.9.3                                                Ques3.sh

#!/bin/bash

awk  -F, 'tolower($1) ~ /^j.*n$/' Data.csv
```

Output:

```
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ nano Data.csv
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ nano Ques3.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ bash Ques3.sh
John,34,Development
```

Question 04:
From ifconfig output find all IP addresses.

Bash Script:

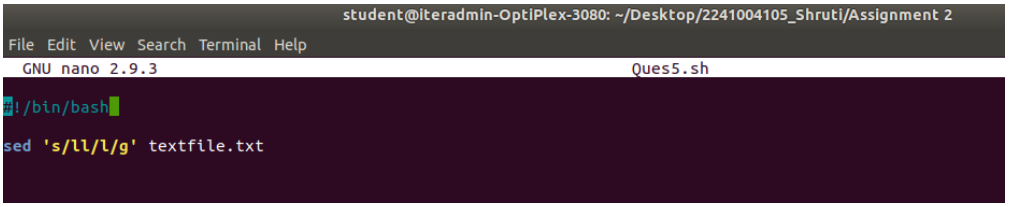
```
                                    student@iteradmin-OptiPlex-3080: ~/Desktop/2241004105_Shruti/Assignment 2
File  Edit  View  Search  Terminal  Help
  GNU nano 2.9.3                                                Ques4.sh

#!/bin/bash
ifconfig | grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}'
```

Output:

```
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ bash Ques4.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ nano Ques4.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ bash Ques4.sh
172.19.0.1
255.255.0.0
172.19.255.255
172.18.0.1
255.255.0.0
172.18.255.255
172.17.131.56
255.255.240.0
172.17.143.255
127.0.0.1
255.0.0.0
192.168.122.1
255.255.255.0
192.168.122.255
```

Question 05:
Replace all occurrences of 'll' with a single 'l'.
Bash Script:



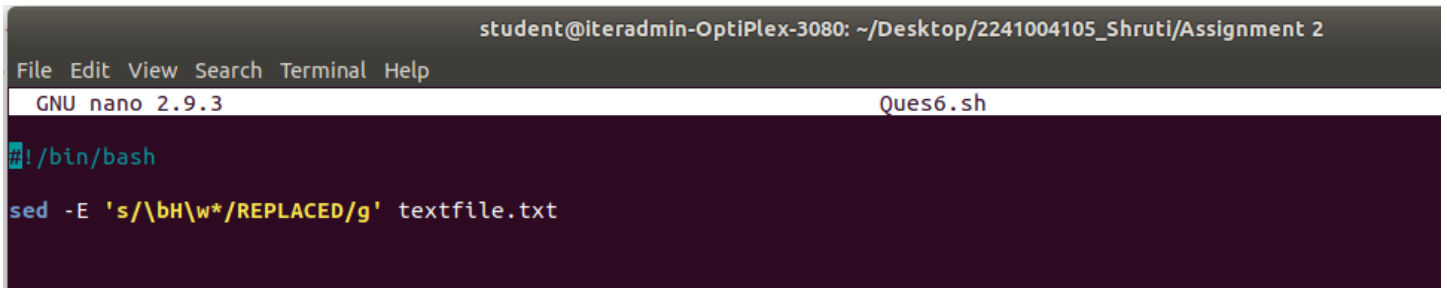
Output:

```
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ nano Ques5.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ bash Ques5.sh
Helo, I am Shruti Das
This is a text
we have to extract TestT100String from this.
let us try to do this
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$
```

Question 06:

Replace a word that starts with H.

Bash Script:



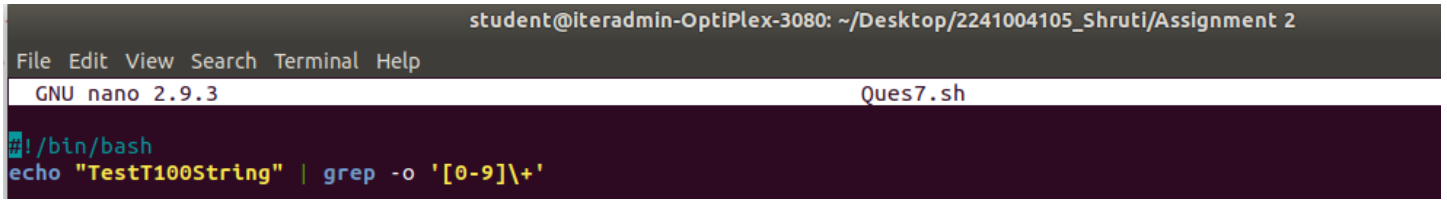
Output:

```
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ nano Ques6.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ bash Ques6.sh
REPLACED, I am Shruti Das
This is a text
we have to extract TestT100String from this.
let us try to do this
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$
```

Question 07:

From the text TestT100String only extract the digits.

Bash Script:



Output:

```
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ nano Ques6.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ nano Ques7.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ bash Ques7.sh
100
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$
```

Question 08:

Write a Bash script to find the latest file in any given folder. The folder should be presented as a command line argument. Make a copy of the file and find the

most occurring word in the file.

Bash Script:

```
student@iteradmin-OptiPlex-3080: ~/Desktop/2241004105_Shruti/Assignment 2
File  Edit  View  Search  Terminal  Help
  GNU nano 2.9.3                                                    Ques8.sh

#!/bin/bash
folder="$1"

latest=$(ls -t "$folder" | head -1)

cp "$folder/$latest" copy.txt

cat copy.txt | tr ' ' '\n' | sort | uniq -c | sort -nr | head -3
```

Output:

```
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ bash Ques8.sh "/home/student/Desktop/2241004105_Shruti/Assignment 2"
      2 you
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ ls -t | head -n1
copy.txt
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ cat copy.txt
The Japanese word ikigai combines "iki" (which means life) and "gai" (value).
It shows that everyone has a purpose in life to offer the world.
It's what brings you joy and inspires you to get out of bed every day.
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ nano Ques8.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ nano New_file.txt
Use "fg" to return to nano.

[3]+  Stopped                 nano New_file.txt
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ bash Ques8.sh "/home/student/Desktop/2241004105_Shruti/Assignment 2"
      2 you
      2 to
      2 and
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$
```

Question 09:

Write a Bash script to print the name of the the person who last logged into the system. From the home folder find out (if any) files for which this person is the owner. Can you find the total login time of this user

Bash Script:

```
student@iteradmin-OptiPlex-3080: ~/Desktop/2241004105_Shruti/Assignment 2
File  Edit  View  Search  Terminal  Help
  GNU nano 2.9.3                                                    Ques9.sh

#!/bin/bash

user=$(last | head -1 | awk '{print $1}')
echo "User: $user"

find /home/student/Desktop -user "$user" 2>/dev/null

last "$user" | head -3
```

Output:

```
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ bash Ques9.sh
User: student
/home/student/Desktop
/home/student/Desktop/day
/home/student/Desktop/2241004105_Shruti
/home/student/Desktop/2241004105_Shruti/files_deleted.txt
/home/student/Desktop/2241004105_Shruti/Ques1.sh
/home/student/Desktop/2241004105_Shruti/Ques3.sh.save
/home/student/Desktop/2241004105_Shruti/Ques5.sh
/home/student/Desktop/2241004105_Shruti/shruti123_credentials.csv
/home/student/Desktop/2241004105_Shruti/Ques4.sh
/home/student/Desktop/2241004105_Shruti/Ques5.sh.save
/home/student/Desktop/2241004105_Shruti/Ques7.sh.save
/home/student/Desktop/2241004105_Shruti/Assignment 1
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques5(Output).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques6(Script).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques1(script).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques9(output).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques3(output).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques10(output).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques2(Output).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques8(script).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques7(csv_file).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques4(output).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques6(Output).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques9(Script).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques4(Script).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques5(script).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques3(Script).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques1(output).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques10(script).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques7(script).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques7(Output).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques8(Output).png
/home/student/Desktop/2241004105_Shruti/Assignment 1/Ques2(script).png
```

Question 10:
Write a Bash script to use find command to find the path of conf.d file. From the path replace all "/" symbols and replace them with "-" and print the path.
Bash Script:

```
student@iteradmin-OptiPlex-3080: ~/Desktop/2241004105_Shruti/Assignment 2
File  Edit  View  Search  Terminal  Help
  GNU nano 2.9.3                                    Ques10.sh

#!/bin/bash

path=$(find /etc -name "conf.d" 2>/dev/null)

echo "$path" | tr '/' '-'
```

Output:

```
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/Assignment 2$ bash Ques10.sh
-etc-NetworkManager-conf.d
-etc-nginx-conf.d
-etc-fonts-conf.d
-etc-initramfs-tools-conf.d
-etc-libblockdev-conf.d
```

AWS ASSIGNMENT
TOPIC: IAM(Identity and Access Management)

Name : Shruti Das
Registration : 2241004105
Date : 03/07/2025 and 04/07/2025

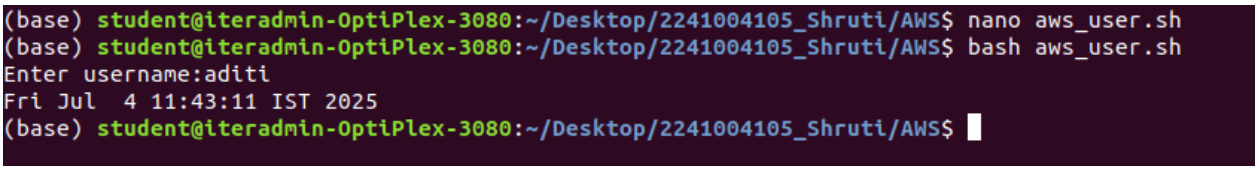
Question 01:
Go through all the CLI commands of IAM.

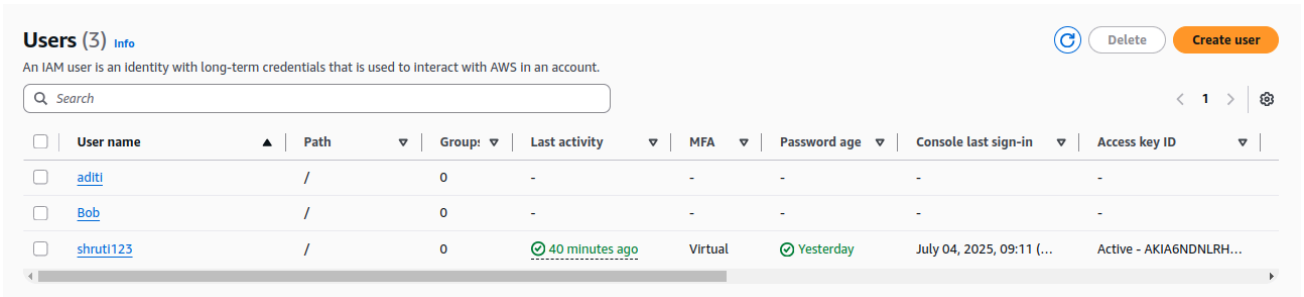
Question 02:
Write a Bash Script to create an aws user in the script. Find out the user id. Also, print the creation time stamp.

Bash Script:



Output:





Question 03:
Write a Bash Script to create an IAM user. Add the following policies: EC2, Administrator, Iam Full access.

student@iteradmin-OptiPlex-3080: ~/Desktop/2241004105_Shruti/AWS

File Edit View Search Terminal Help

GNU nano 2.9.3 attach_policy.sh

```
#!/bin/bash
read -p "Enter username:" USERNAME
aws iam create-user --user-name "$USERNAME" | grep '"User Id"' | cut -d '"' -f4

aws iam attach-user-policy \
    --policy-arn arn:aws:iam::aws:policy/AdministratorAccess \
    --user-name $USERNAME
aws iam attach-user-policy \
    --policy-arn arn:aws:iam::aws:policy/AmazonEC2FullAccess \
    --user-name $USERNAME
aws iam attach-user-policy \
    --policy-arn arn:aws:iam::aws:policy/IAMFullAccess \
    --user-name $USERNAME
```

```
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/AWS$ nano aws_user.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/AWS$ nano attach_policy.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/AWS$ bash attach_policy.sh
Enter username:alice
```

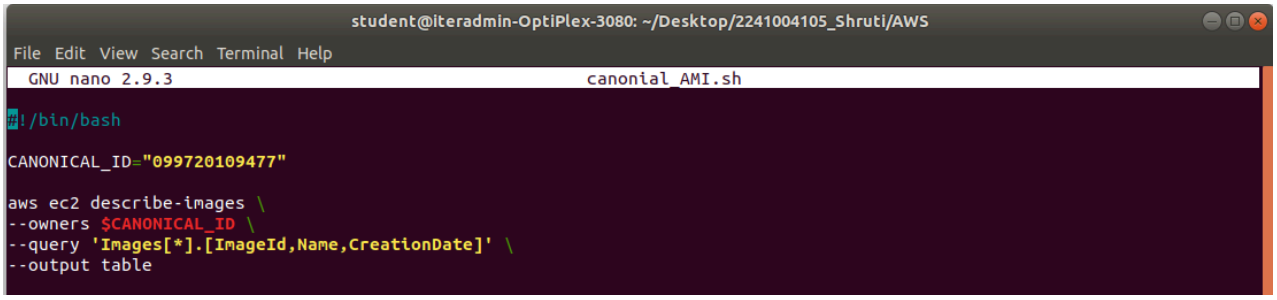
User name	Path	Groups	Last activity	MFA	Password age	Console last sign-in	Access key ID
aditi	/	0	-	-	-	-	-
alice	/	0	-	-	-	-	-
Bob	/	0	-	-	-	-	-
shruti123	/	0	16 minutes ago	Virtual	Yesterday	July 04, 2025, 09:11 (...	Active - AKIA6NDNLRH...

AWS ASSIGNMENT
TOPIC: EC2

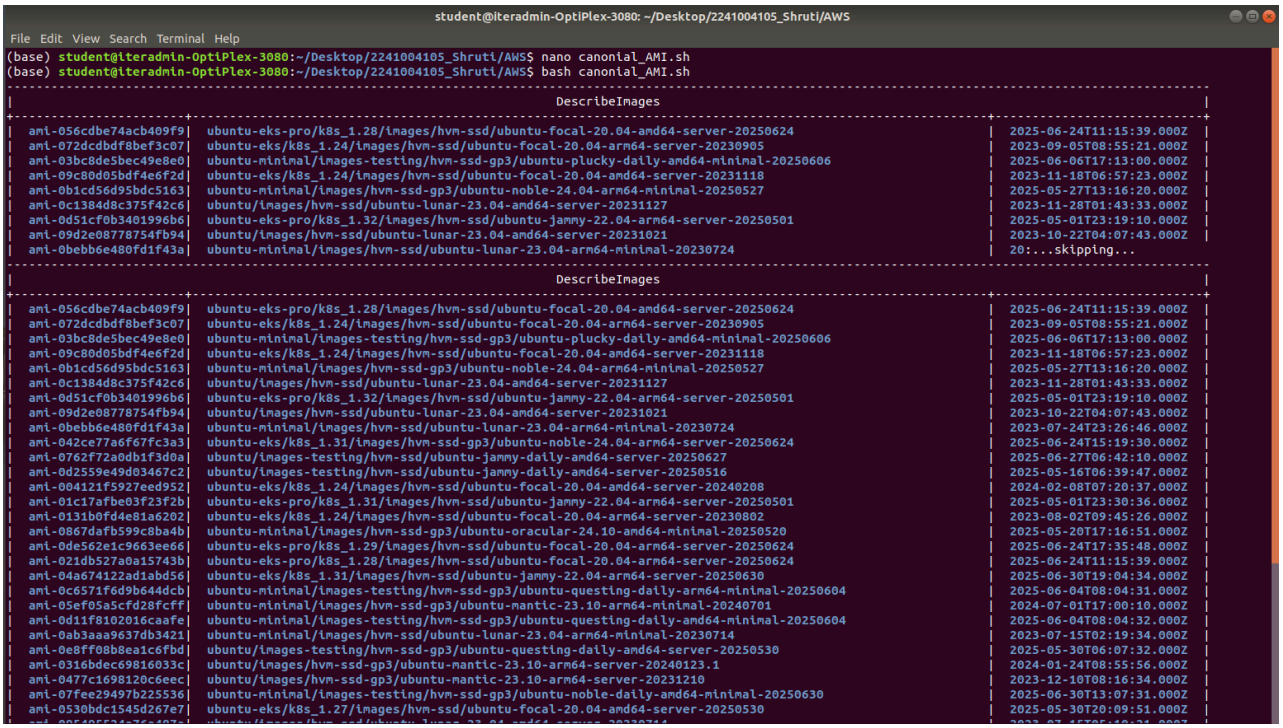
Name : Shruti Das
Registration : 2241004105
Date : 03/07/2025 and 04/07/2025

Question 01:
Write a bash script to find all the available AMIs from the AMI store from the list print only those AMIs which are made by canonical.

Bash Script:



Output:


student@iteradmin-OptiPlex-3080: ~/Desktop/2241004105_Shruti/AWS
File Edit View Search Terminal Help
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/AWS$ nano canonial_AMI.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/AWS$ bash canonial_AMI.sh
DescribeImages
ami-056cdbe74acb409f9	ubuntu-eks-pro/k8s_1.28/images/hvm-ssd/ubuntu-focal-20.04-amd64-server-20250624	2025-06-24T11:15:39.000Z
ami-072dcdbdf8bef3c07	ubuntu-eks/k8s_1.24/images/hvm-ssd/ubuntu-focal-20.04-arm64-server-20230905	2023-09-05T08:55:21.000Z
ami-03bc8de5bec49e8e0	ubuntu-minimal/images-testing/hvm-ssd-gp3/ubuntu-plucky-daily-amd64-minimal-20250606	2025-06-06T17:13:00.000Z
ami-09c80d05bdf4e6f2d	ubuntu-eks/k8s_1.24/images/hvm-ssd/ubuntu-focal-20.04-amd64-server-20231118	2023-11-18T06:57:23.000Z
ami-0b1cd56d95bdc5163	ubuntu-minimal/images/hvm-ssd-gp3/ubuntu-noble-24.04-arm64-minimal-20250527	2025-05-27T13:16:20.000Z
ami-0c1384d8c375f42c6	ubuntu/images/hvm-ssd/ubuntu-lunar-23.04-amd64-server-20231127	2023-11-28T01:43:33.000Z
ami-0d51cf0b3401996b6	ubuntu-eks-pro/k8s_1.32/images/hvm-ssd/ubuntu-jammy-22.04-arm64-server-20250501	2025-05-01T23:19:10.000Z
ami-09d2e08778754fb94	ubuntu/images/hvm-ssd/ubuntu-lunar-23.04-amd64-server-20231021	2023-10-22T04:07:43.000Z
ami-0bebb6e480fd1f43a	ubuntu-minimal/images/hvm-ssd/ubuntu-lunar-23.04-arm64-minimal-20230724	20:...skipping...
DescribeImages
ami-056cdbe74acb409f9	ubuntu-eks-pro/k8s_1.28/images/hvm-ssd/ubuntu-focal-20.04-amd64-server-20250624	2025-06-24T11:15:39.000Z
ami-072dcdbdf8bef3c07	ubuntu-eks/k8s_1.24/images/hvm-ssd/ubuntu-focal-20.04-arm64-server-20230905	2023-09-05T08:55:21.000Z
ami-03bc8de5bec49e8e0	ubuntu-minimal/images-testing/hvm-ssd-gp3/ubuntu-plucky-daily-amd64-minimal-20250606	2025-06-06T17:13:00.000Z
ami-09c80d05bdf4e6f2d	ubuntu-eks/k8s_1.24/images/hvm-ssd/ubuntu-focal-20.04-amd64-server-20231118	2023-11-18T06:57:23.000Z
ami-0b1cd56d95bdc5163	ubuntu-minimal/images/hvm-ssd-gp3/ubuntu-noble-24.04-arm64-minimal-20250527	2025-05-27T13:16:20.000Z
ami-0c1384d8c375f42c6	ubuntu/images/hvm-ssd/ubuntu-lunar-23.04-amd64-server-20231127	2023-11-28T01:43:33.000Z
ami-0d51cf0b3401996b6	ubuntu-eks-pro/k8s_1.32/images/hvm-ssd/ubuntu-jammy-22.04-arm64-server-20250501	2025-05-01T23:19:10.000Z
ami-09d2e08778754fb94	ubuntu/images/hvm-ssd/ubuntu-lunar-23.04-amd64-server-20231021	2023-10-22T04:07:43.000Z
ami-0bebb6e480fd1f43a	ubuntu-minimal/images/hvm-ssd/ubuntu-lunar-23.04-arm64-minimal-20230724	2023-07-24T23:26:46.000Z
ami-042ce77a6f67fc3a3	ubuntu-eks/k8s_1.31/images/hvm-ssd-gp3/ubuntu-noble-24.04-arm64-server-20250624	2025-06-24T15:19:30.000Z
ami-0762f72a0db1f3d0a	ubuntu/images-testing/hvm-ssd/ubuntu-jammy-daily-amd64-server-20250627	2025-06-27T06:42:10.000Z
ami-0d2559e49d03467c2	ubuntu/images-testing/hvm-ssd/ubuntu-jammy-daily-amd64-server-20250516	2025-05-16T06:39:47.000Z
ami-004121f5927eed952	ubuntu-eks/k8s_1.24/images/hvm-ssd/ubuntu-focal-20.04-amd64-server-20240208	2024-02-08T07:20:37.000Z
ami-01c17afbe03f23f2b	ubuntu-eks-pro/k8s_1.31/images/hvm-ssd/ubuntu-jammy-22.04-arm64-server-20250501	2025-05-01T23:30:36.000Z
ami-0131b0fd4e81a6202	ubuntu-eks/k8s_1.24/images/hvm-ssd/ubuntu-focal-20.04-arm64-server-20230802	2023-08-02T09:45:26.000Z
ami-0867dafb599c8ba4b	ubuntu-minimal/images/hvm-ssd-gp3/ubuntu-oracular-24.10-amd64-minimal-20250520	2025-05-20T17:16:51.000Z
ami-0de562e1c9663ee66	ubuntu-eks-pro/k8s_1.29/images/hvm-ssd/ubuntu-focal-20.04-arm64-server-20250624	2025-06-24T17:35:48.000Z
ami-021db527a0a15743b	ubuntu-eks-pro/k8s_1.28/images/hvm-ssd/ubuntu-focal-20.04-arm64-server-20250624	2025-06-24T11:15:39.000Z
ami-04a674122ad1abd56	ubuntu-eks/k8s_1.31/images/hvm-ssd/ubuntu-jammy-22.04-arm64-server-20250630	2025-06-30T19:04:34.000Z
ami-0c6571f6d9b644dcb	ubuntu-minimal/images-testing/hvm-ssd-gp3/ubuntu-questing-daily-arm64-minimal-20250604	2025-06-04T08:04:31.000Z
ami-05ef05a5cfd28fcff	ubuntu-minimal/images/hvm-ssd-gp3/ubuntu-mantic-23.10-arm64-minimal-20240701	2024-07-01T17:00:10.000Z
ami-0d11f8102016caafe	ubuntu-minimal/images-testing/hvm-ssd-gp3/ubuntu-questing-daily-amd64-minimal-20250604	2025-06-04T08:04:32.000Z
ami-0ab3aaa9637db3421	ubuntu-minimal/images/hvm-ssd/ubuntu-lunar-23.04-arm64-minimal-20230714	2023-07-15T02:19:34.000Z
ami-0e8ff08b8ea1c6fbd	ubuntu/images-testing/hvm-ssd-gp3/ubuntu-questing-daily-amd64-server-20250530	2025-05-30T06:07:32.000Z
ami-0316bdec69816033c	ubuntu/images/hvm-ssd-gp3/ubuntu-mantic-23.10-arm64-server-20240123.1	2024-01-24T08:55:56.000Z
ami-0477c1698120c6eec	ubuntu/images/hvm-ssd-gp3/ubuntu-mantic-23.10-arm64-server-20231210	2023-12-10T08:16:34.000Z
ami-07fee29497b225536	ubuntu-minimal/images-testing/hvm-ssd-gp3/ubuntu-noble-daily-amd64-minimal-20250630	2025-06-30T13:07:31.000Z
ami-0530bdc1545d267e7	ubuntu-eks/k8s_1.27/images/hvm-ssd/ubuntu-focal-20.04-amd64-server-20250530	2025-05-30T20:09:51.000Z

Question 02:
Write a Bash Script to create an Instance using the ubuntu free tier AMI. The script should create a key pair and attach it to an Instance, it must also attach to a Security group where SSH is allowed from anywhere and is allowed for my ip. Once the Instance is created the script must print the Instance id only.
Question 03:
After creating the Instance, also print the DNS of the Instance.

```bash
#!/bin/bash
set -e

# Variables
KEY_NAME="my-ubuntu-key-$(date +%s)"
SEC_GROUP_NAME="my-ssh-sg-$(date +%s)"
REGION="eu-north-1"
INSTANCE_TYPE="t3.micro"

# Fetch your public IP
MY_IP=$(curl -s https://checkip.amazonaws.com)/32
echo "Your IP: $MY_IP"

# Create a key pair
echo "Creating key pair: $KEY_NAME"
aws ec2 create-key-pair --key-name "$KEY_NAME" --query 'KeyMaterial' --output text > "${KEY_NAME}.pem"
chmod 400 "${KEY_NAME}.pem"

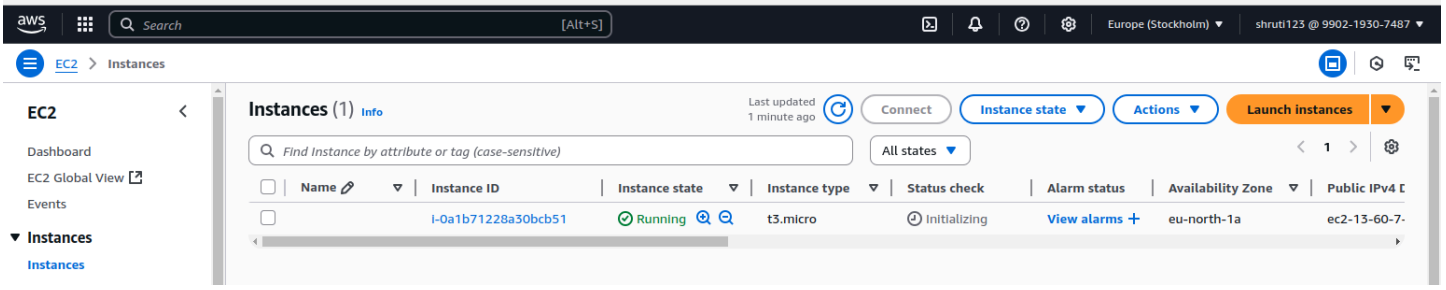
# Create a security group
echo "Creating security group: $SEC_GROUP_NAME"
SEC_GROUP_ID=$(aws ec2 create-security-group --group-name "$SEC_GROUP_NAME" --description "Allow SSH from anywhere and my IP" --query 'GroupId' --output text --region $$

# Authorize inbound SSH from anywhere
echo "Authorizing inbound SSH from anywhere (0.0.0.0/0)..."
aws ec2 authorize-security-group-ingress --group-id "$SEC_GROUP_ID" --protocol tcp --port 22 --cidr 0.0.0.0/0 --region $REGION

# Authorize inbound SSH from your IP
echo "Authorizing inbound SSH from your IP ($MY_IP)..."
aws ec2 authorize-security-group-ingress --group-id "$SEC_GROUP_ID" --protocol tcp --port 22 --cidr "$MY_IP" --region $REGION

# Retrieve the latest Ubuntu 22.04 LTS AMI ID using SSM
echo "Fetching the latest Ubuntu 22.04 LTS AMI ID..."
AMI_ID=$(aws ssm get-parameters --names "/aws/service/canonical/ubuntu/server/jammy/stable/current/amd64/hvm/ebs-gp2/ami-id" --query "Parameters[0].Value" --output text$

# Launch the EC2 instance
echo "Launching EC2 instance..."
INSTANCE_ID=$(aws ec2 run-instances \
  --image-id "$AMI_ID" \
  --count 1 \
  --instance-type "$INSTANCE_TYPE" \
```




```
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/AWS$ nano ec2_ques2.sh
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/AWS$ bash ec2_ques2.sh
Your IP: 115.240.101.53/32
Creating key pair: my-ubuntu-key-1752561568
Creating security group: my-ssh-sg-1752561568
Authorizing inbound SSH from anywhere (0.0.0.0/0)...
{
    "Return": true,
    "SecurityGroupRules": [
        {
            "SecurityGroupRuleId": "sgr-0c721bc53f9646368",
            "GroupId": "sg-082ebda55a283771b",
            "GroupOwnerId": "990219307487",
            "IsEgress": false,
            "IpProtocol": "tcp",
            "FromPort": 22,
            "ToPort": 22,
            "CidrIpv4": "0.0.0.0/0"
        }
    ]
}
Authorizing inbound SSH from your IP (115.240.101.53/32)...
{
    "Return": true,
    "SecurityGroupRules": [
        {
            "SecurityGroupRuleId": "sgr-02d1ec1707f2d518e",
            "GroupId": "sg-082ebda55a283771b",
            "GroupOwnerId": "990219307487",
            "IsEgress": false,
            "IpProtocol": "tcp",
            "FromPort": 22,
            "ToPort": 22,
            "CidrIpv4": "115.240.101.53/32"
        }
    ]
}
Fetching the latest Ubuntu 22.04 LTS AMI ID...
Launching EC2 instance...
Instance created with ID: i-0a1b71228a30bcb51
Waiting for instance to be in 'running' state...
Instance Public DNS: ec2-13-60-7-44.eu-north-1.compute.amazonaws.com
```

Question 04:
Go through all the EC2 Commands.

Question 05:
Find out all the options and parameters of SSH from man or help

Question 06:

Write a bash script to create a t2.micro instance and add a EBS volume to it. In the same script, terminate the earlier Instance and create a new t2.micro Instance. Attach the EBS Volume to the new Instance and display the instance ID of the new Instance.

```bash
#!/bin/bash

REGION="eu-north-1"
INSTANCE_TYPE="t3.micro"
KEY_NAME="my-ubuntu-key-1752561454"
SECURITY_GROUP_ID="sg-0504d6f30ed975ef0"


AMI=$(aws ec2 describe-images \
  --owners 099720109477 \
  --filters "Name=name,Values=ubuntu/images/hvm-ssd/ubuntu-jammy-22.04-amd64-server-*" \
  --query 'Images | sort_by(@, &CreationDate) | [-1].ImageId' \
  --output text --region $REGION)

echo "Using AMI: $AMI"

INSTANCE1=$(aws ec2 run-instances \
  --image-id $AMI \
  --instance-type $INSTANCE_TYPE \
  --key-name $KEY_NAME \
  --security-group-ids $SECURITY_GROUP_ID \
  --query 'Instances[0].InstanceId' \
  --output text --region $REGION)

echo "First instance launched: $INSTANCE1"

aws ec2 wait instance-running --instance-ids $INSTANCE1 --region $REGION

AZ=$(aws ec2 describe-instances --instance-ids $INSTANCE1 --query 'Reservations[0].Instances[0].Placement.AvailabilityZone' --output text --region $REGION)

VOLUME=$(aws ec2 create-volume --size 10 --availability-zone $AZ --query 'VolumeId' --output text --region $REGION)

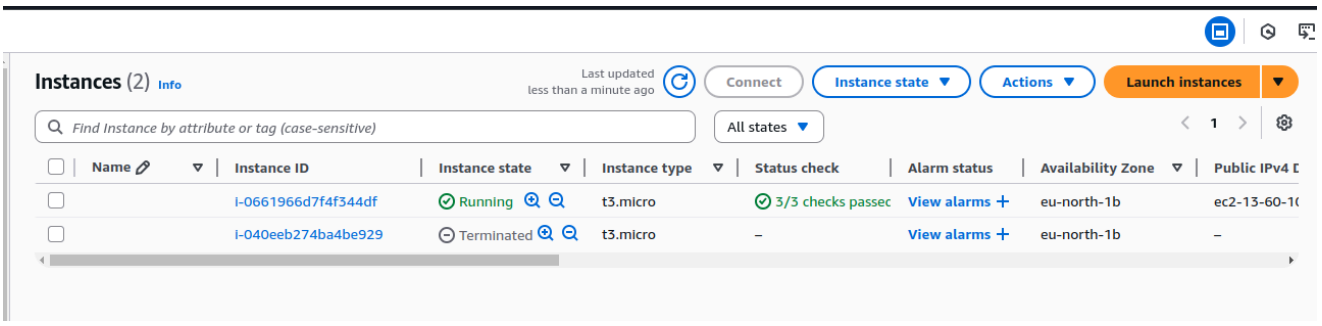
echo "Created volume: $VOLUME"

aws ec2 wait volume-available --volume-ids $VOLUME --region $REGION

aws ec2 attach-volume --volume-id $VOLUME --instance-id $INSTANCE1 --device /dev/sdf --region $REGION

echo "Volume attached to first instance."
```

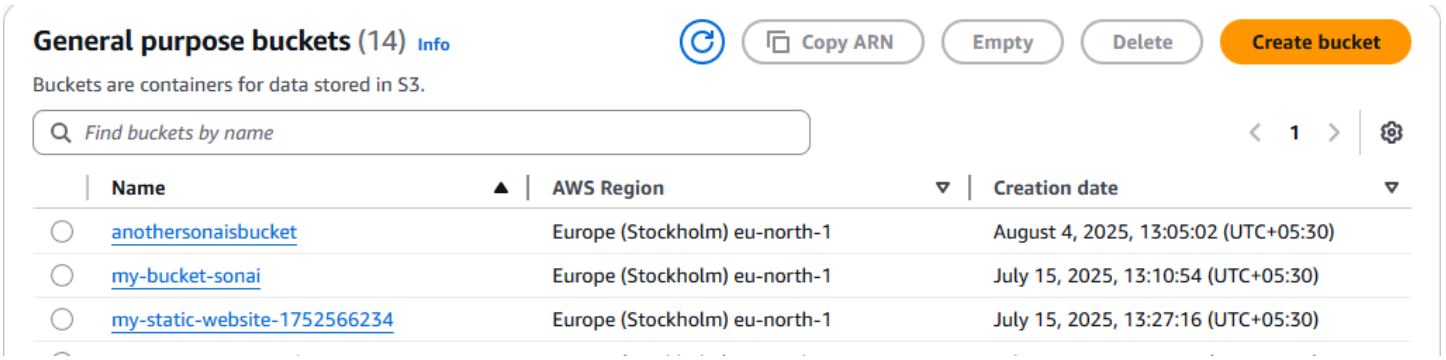
```
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/AWS$ bash ec2_Ques6.sh
Using AMI: ami-0b8e4d801c75b0f0d
First instance launched: i-040eeb274ba4be929
Created volume: vol-0232507ee1300a1ea
{
    "AttachTime": "2025-07-16T09:27:57.884000+00:00",
    "Device": "/dev/sdf",
    "InstanceId": "i-040eeb274ba4be929",
    "State": "attaching",
    "VolumeId": "vol-0232507ee1300a1ea"
}
Volume attached to first instance.
{
    "TerminatingInstances": [
        {
            "CurrentState": {
                "Code": 32,
                "Name": "shutting-down"
            },
            "InstanceId": "i-040eeb274ba4be929",
            "PreviousState": {
                "Code": 16,
                "Name": "running"
            }
        }
    ]
}
Second instance launched: i-0661966d7f4f344df
{
    "AttachTime": "2025-07-16T09:29:23.784000+00:00",
    "Device": "/dev/sdf",
    "InstanceId": "i-0661966d7f4f344df",
    "State": "attaching",
    "VolumeId": "vol-0232507ee1300a1ea"
}
i-0661966d7f4f344df
```

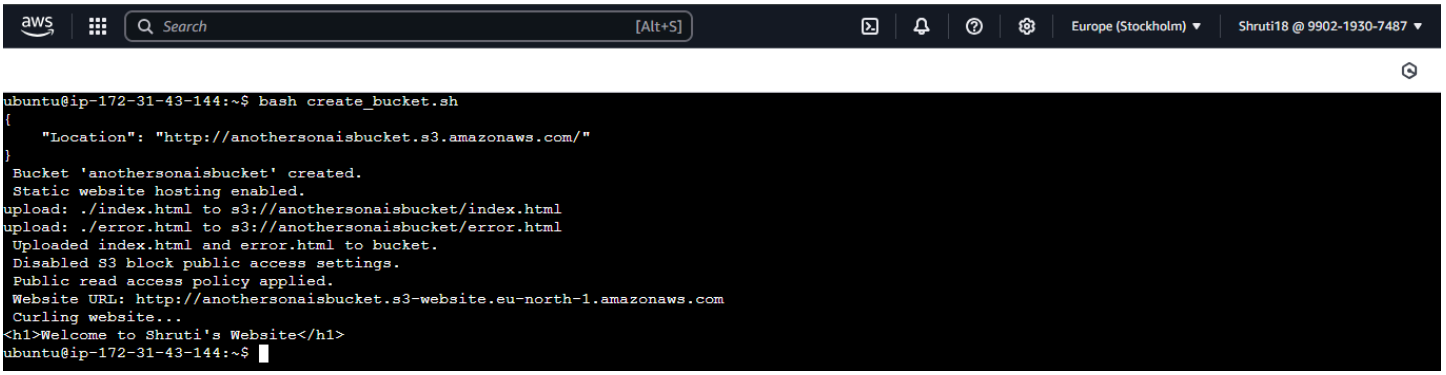


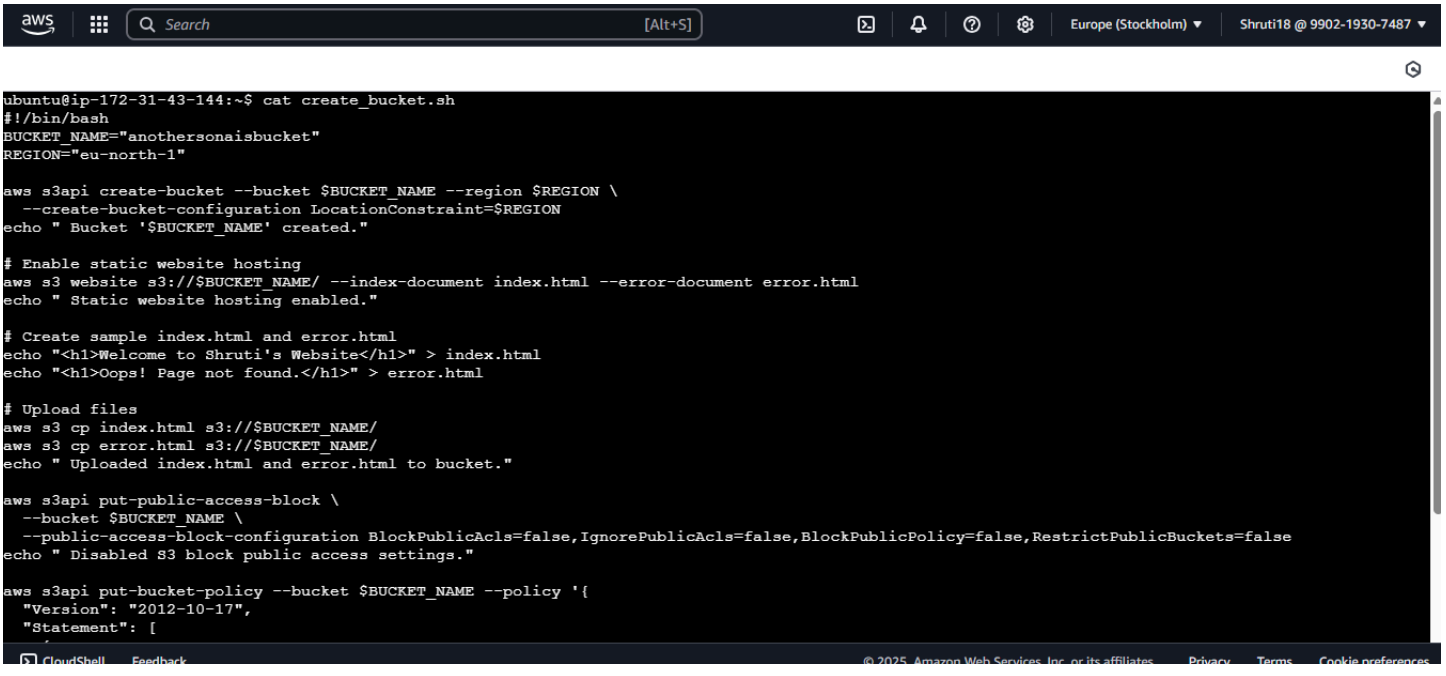
AWS ASSIGNMENT
TOPIC:S3

Name: Shruti Das
Registration no.: 2241004105
Date: 07/07/2025

Question 01:
Write a bash script to create a bucket and configure it as a Website. The script should also upload index.html and error.html to the bucket, then it should show the url of the bucket and script should also curl that url.







Question 02:
Go through all the commands of S3.

Question 03:
Write a bash script to create a bucket and encrypt the bucket.

```
#!/bin/bash
BUCKET_NAME=$1
REGION=$2

if [ -z "$BUCKET_NAME" ] || [ -z "$REGION" ]; then
  echo "Usage: $0 <bucket-name> <region>"
  exit 1
fi

aws s3api create-bucket --bucket "$BUCKET_NAME" --region "$REGION" \
  --create-bucket-configuration LocationConstraint="$REGION"

if [ $? -ne 0 ]; then
  echo "Bucket creation failed."
  exit 1
fi

echo "Bucket $BUCKET_NAME created successfully."

# Enable default encryption with AWS-managed keys (SSE-S3)
aws s3api put-bucket-encryption --bucket "$BUCKET_NAME" --server-side-encryption-configuration '{
  "Rules": [
    {
      "ApplyServerSideEncryptionByDefault": {
        "SSEAlgorithm": "AES256"
      }
    }
  ]
}'

if [ $? -ne 0 ]; then
  echo "Failed to enable encryption on bucket."
  exit 1
fi

echo "Encryption enabled on bucket $BUCKET_NAME."
```

```
(base) student@iteradmin-OptiPlex-3080:~/Desktop/2241004105_Shruti/AWS$ bash s3_Ques2.sh my-bucket-sonai eu-north-1
{
    "Location": "http://my-bucket-sonai.s3.amazonaws.com/"
}
Bucket my-bucket-sonai created successfully.
Encryption enabled on bucket my-bucket-sonai.
```

Question 04:
Write a bash script to create a s3 bucket. The bucket name should be created using a random function where you have no idea of the name of the bucket. Use some commands to retrieve the bucket name and cp a script file to it. Display the contents of the bucket and delete it.

```
File  Edit  View  Search  Terminal  Help
  GNU nano 2.9.3
```

```
#!/bin/bash
BUCKET_NAME="demo-bucket-$(date +%s)-$RANDOM"

aws s3 mb s3://$BUCKET_NAME
echo "created: $BUCKET_NAME"

echo -e "#!/bin/bash\necho hello!" > upload.sh
chmod +x upload.sh

aws s3 cp upload.sh s3://$BUCKET_NAME/
echo "Uploaded script"

echo "contents of bucket is:"
aws s3 ls s3://$BUCKET_NAME

aws s3 rm s3://$BUCKET_NAME/ --recursive

aws s3 rb s3://$BUCKET_NAME

echo "Bucket deleted with its contents"
```

```
(base) student@iteradmin-OptiPlex-3080:~$ nano Ques1_bucket.sh
(base) student@iteradmin-OptiPlex-3080:~$ nano Ques1_bucket.sh
(base) student@iteradmin-OptiPlex-3080:~$ bash Ques1_bucket.sh
make_bucket: demo-bucket-1752210923-12995
created: demo-bucket-1752210923-12995
upload: ./upload.sh to s3://demo-bucket-1752210923-12995/upload.sh
Uploaded script
contents of bucket is:
2025-07-11 10:45:27         24 upload.sh
delete: s3://demo-bucket-1752210923-12995/upload.sh
remove_bucket: demo-bucket-1752210923-12995
Bucket deleted with its contents
(base) student@iteradmin-OptiPlex-3080:~$
```

Question 05:

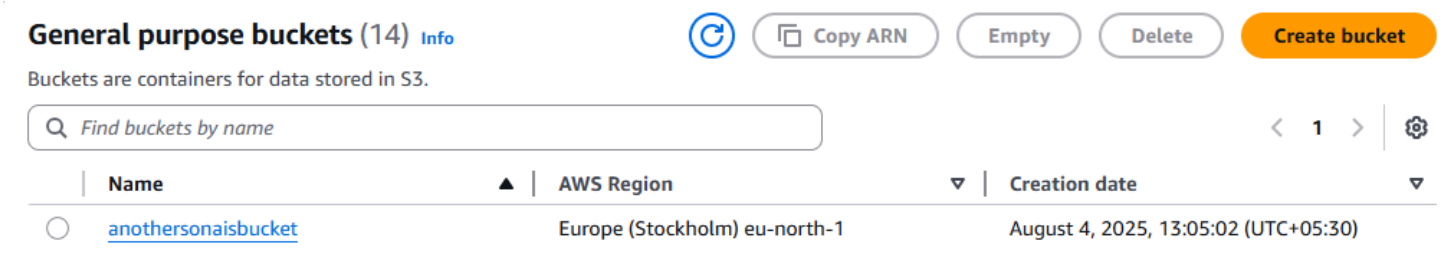
Write a bash script to modify the access control policy of an existing bucket.

```
ubuntu@ip-172-31-43-144:~$ nano modify_bucket.sh
ubuntu@ip-172-31-43-144:~$ bash modify_bucket.sh
 Bucket policy for 'anothersonaisbucket' modified to allow access only from IP range 103.21.244.0/22
ubuntu@ip-172-31-43-144:~$
```

```
ubuntu@ip-172-31-43-144:~$ cat modify_bucket.sh
#!/bin/bash

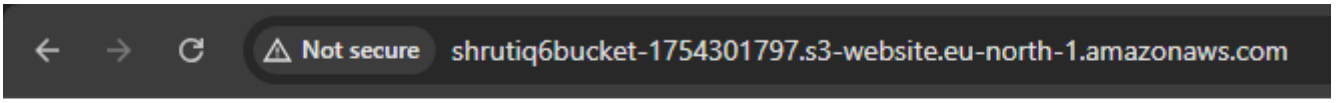
# Variables
BUCKET_NAME="anothersonaisbucket"

# Apply a new bucket policy (example: allow public read access)
aws s3api put-bucket-policy --bucket $BUCKET_NAME --policy '{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowGetFromSpecificIP",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::'"$BUCKET_NAME"'/*",
      "Condition": {
        "IpAddress": {
          "aws:SourceIp": "103.21.244.0/22"
        }
      }
    }
  ]
}'
echo " Bucket policy for '$BUCKET_NAME' modified to allow access only from IP range 103.21.244.0/22"
ubuntu@ip-172-31-43-144:~$
```



Question 06:

Write a bash script to create a bucket and host a static website in it. Is the website accessible through curl and through a browser?



Hello from Shruti: Question 6 ongoing




```
ubuntu@ip-172-31-43-144:~$ nano Ques6.sh
ubuntu@ip-172-31-43-144:~$ bash Ques6.sh
{
    "Location": "http://shrutiq6bucket-1754301797.s3.amazonaws.com/"
}
 Bucket 'shrutiq6bucket-1754301797' created.
 Static website hosting enabled.
upload: ./index.html to s3://shrutiq6bucket-1754301797/index.html
upload: ./error.html to s3://shrutiq6bucket-1754301797/error.html
 HTML files uploaded.
🔓Public access unblocked.
 Public-read policy applied.
 Website URL: http://shrutiq6bucket-1754301797.s3-website.eu-north-1.amazonaws.com
 Curling website content:
<h1>Hello from Shruti: Question 6 ongoing</h1>

 Open the URL in your browser to check: http://shrutiq6bucket-1754301797.s3-website.eu-north-1.amazonaws.com
ubuntu@ip-172-31-43-144:~$
```




```
ubuntu@ip-172-31-43-144:~$ cat Ques6.sh
#!/bin/bash

# Unique bucket name with timestamp
BUCKET_NAME="shrutiq6bucket-$(date +%s)"
REGION="eu-north-1"

# Create S3 bucket
aws s3api create-bucket --bucket $BUCKET_NAME --region $REGION \
  --create-bucket-configuration LocationConstraint=$REGION
echo " Bucket '$BUCKET_NAME' created."

aws s3 website s3://$BUCKET_NAME/ --index-document index.html --error-document error.html
echo " Static website hosting enabled."


echo "<h1>Hello from Shruti: Question 6 ongoing</h1>" > index.html
echo "<h1>Error Page:Something went wrong!</h1>" > error.html

aws s3 cp index.html s3://$BUCKET_NAME/
aws s3 cp error.html s3://$BUCKET_NAME/
echo " HTML files uploaded."

aws s3api put-public-access-block \
  --bucket $BUCKET_NAME \
  --public-access-block-configuration BlockPublicAcls=false,IgnorePublicAcls=false,BlockPublicPolicy=false,RestrictPublicBuckets=false
echo "🔓Public access unblocked."

aws s3api put-bucket-policy --bucket $BUCKET_NAME --policy '{
  "Version": "2012-10-17",
```

Question 07:

Using AWS CLI creates an Ec2 instance. Use some command to display the publicdns address of the ec2 instance. Using the publicdns ssh into the instance through the script. Write another remote script that should be copied to the ec2 instance. This script should contain steps to create a bucket and attach an access point to it.

```
ubuntu@ip-172-31-43-144:~$ nano Ques7.sh
ubuntu@ip-172-31-43-144:~$ bash Ques7.sh
✅EC2 Instance created: i-0bb589e54ce63ce73
⏳Waiting for instance to be in 'running' state...
🌐Public DNS: ec2-51-20-64-113.eu-north-1.compute.amazonaws.com
ubuntu@ip-172-31-43-144:~$
```

```
  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   40.7% of 6.71GB   Processes:             117
  Memory usage: 32%               Users logged in:       1
  Swap usage:   0%                IPv4 address for ens5: 172.31.43.144

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

   https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


*** System restart required ***
Connected to EC2
ip-172-31-43-144
Remote script executed. Visit: http://ec2-13-51-169-254.eu-north-1.compute.amazonaws.com
ubuntu@ip-172-31-43-144:~$
```

```
ubuntu@ip-172-31-43-144:~$ nano Ques7.sh
ubuntu@ip-172-31-43-144:~$ cat Ques7.sh
#!/bin/bash

# Step 1: Create EC2 instance
INSTANCE_ID=$(aws ec2 run-instances \
  --image-id ami-0014ce3e52359afbd \
  --instance-type t3.micro \
  --key-name Shruti-safekey \
  --security-groups default \
  --region eu-north-1 \
  --query "Instances[0].InstanceId" \
  --output text)

echo " EC2 Instance created: $INSTANCE_ID"

# Step 2: Wait for the instance to be in 'running' state
echo " Waiting for instance to be in 'running' state..."
aws ec2 wait instance-running \
  --instance-ids $INSTANCE_ID \
  --region eu-north-1

# Step 3: Get the public DNS of the instance
DNS=$(aws ec2 describe-instances \
  --instance-ids $INSTANCE_ID \
  --region eu-north-1 \
  --query "Reservations[0].Instances[0].PublicDnsName" \
  --output text)
```

```
ubuntu@ip-172-31-43-144:~$ cat run_ssh.sh
#!/bin/bash

# Replace this with the actual DNS from the AWS Console
PUBLIC_DNS="ec2-13-51-169-254.eu-north-1.compute.amazonaws.com"

# Ensure .pem file exists in the same directory
chmod 400 Shruti-safekey.pem

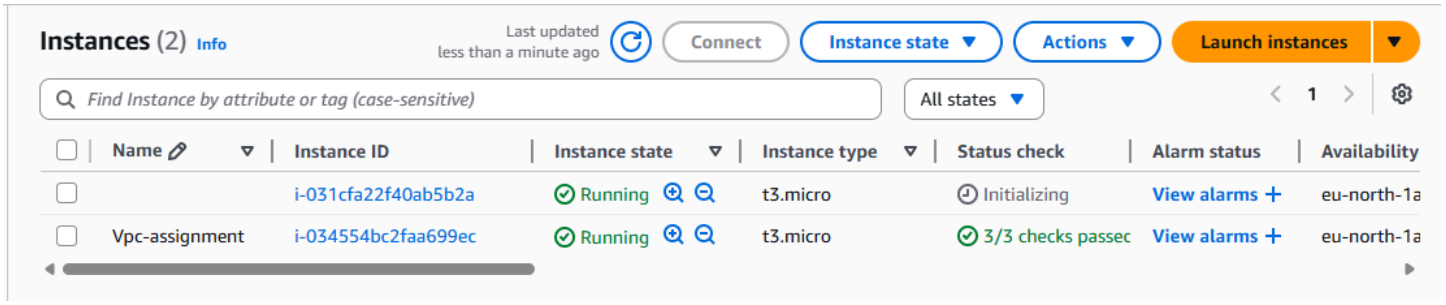
# SSH into the instance
ssh -o StrictHostKeyChecking=no -i Shruti-safekey.pem ubuntu@$PUBLIC_DNS << 'EOF'
echo "Connected to EC2"
hostname
EOF

echo "Remote script executed. Visit: http://$PUBLIC_DNS"
ubuntu@ip-172-31-43-144:~$
```

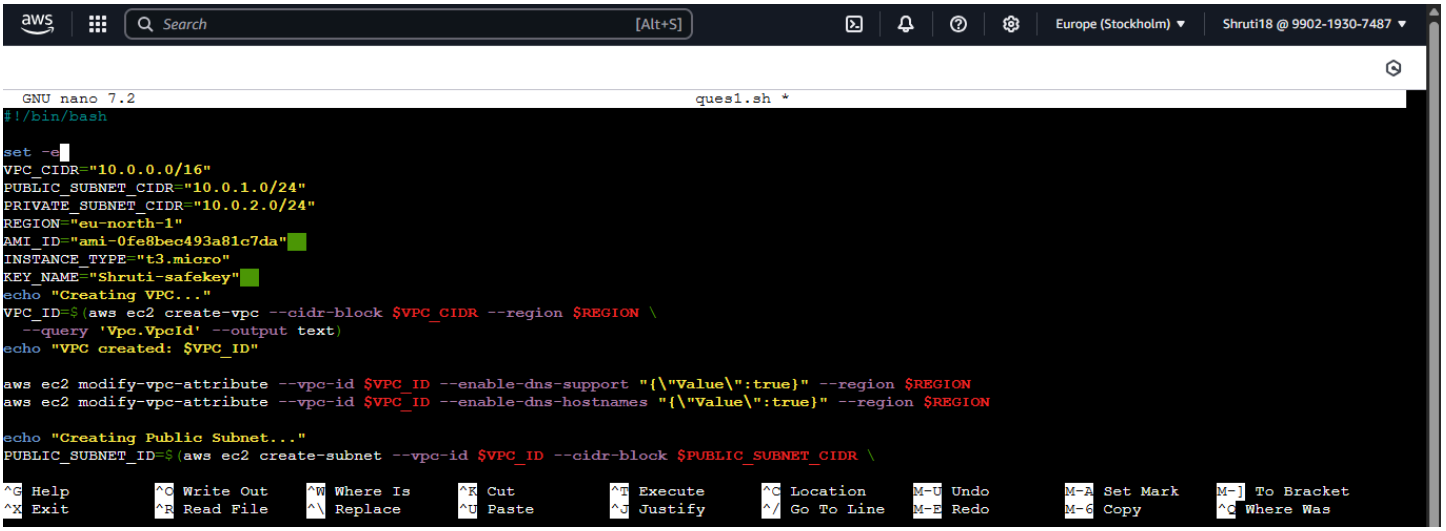
AWS ASSIGNMENT
TOPIC:VPC

Name: Shruti Das
Registration no.: 2241004105
Date: 15/07/2025

Question 01:
Write a Bash Script to create a VPC, create 2 subnets inside the VPC. Create an Internet Gateway. Attach the Internet Gateway to VPC. Create a routing table and attach it to the public subnet. Edit routes and open a route with destination 0.0.0.0/0 and attach it to the internet Gateway. Once this infrastructure is made, launch an Instance into the public subnet. Verify that you are able to SSH. If it does not SSH. Everything fails.




```
            "FromPort": 22,
            "ToPort": 22,
            "CidrIpv4": "0.0.0.0/0",
            "SecurityGroupRuleArn": "arn:aws:ec2:eu-north-1:990219307487:security-group-rule/sgr-083393e6691212820"
        }
    ]
}
{
    "Return": true,
    "SecurityGroupRules": [
        {
            "SecurityGroupRuleId": "sgr-0d6d0abb258788df6",
            "GroupId": "sg-072e39efb78159546",
            "GroupOwnerId": "990219307487",
            "IsEgress": false,
            "IpProtocol": "tcp",
            "FromPort": 80,
            "ToPort": 80,
            "CidrIpv4": "0.0.0.0/0",
            "SecurityGroupRuleArn": "arn:aws:ec2:eu-north-1:990219307487:security-group-rule/sgr-0d6d0abb258788df6"
        }
    ]
}
Launching EC2 instance...
Instance launched: i-031cfa22f40ab5b2a
Waiting for instance to be in 'running' state...
Instance Public IP: 13.53.170.209
 Infrastructure created successfully!
To SSH: ssh -i Shruti-safekey.pem ubuntu@13.53.170.209
```

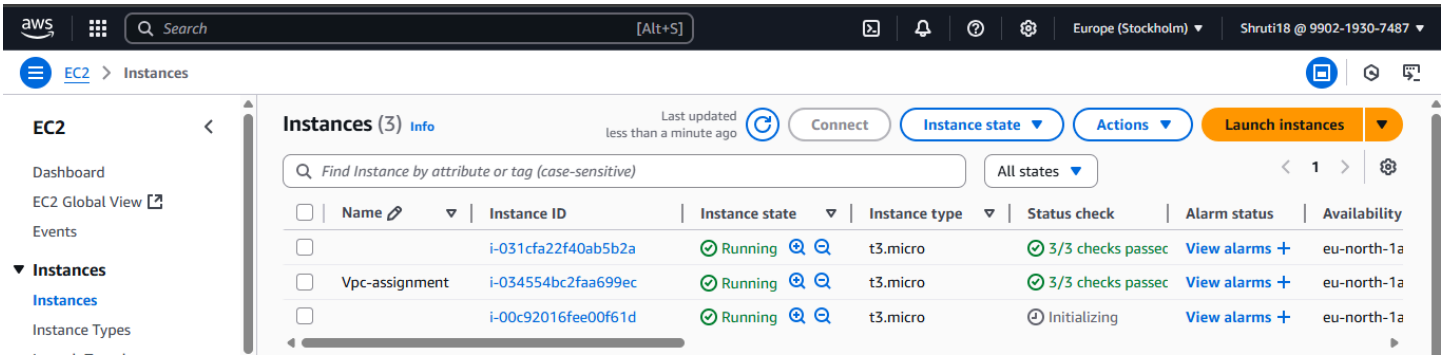


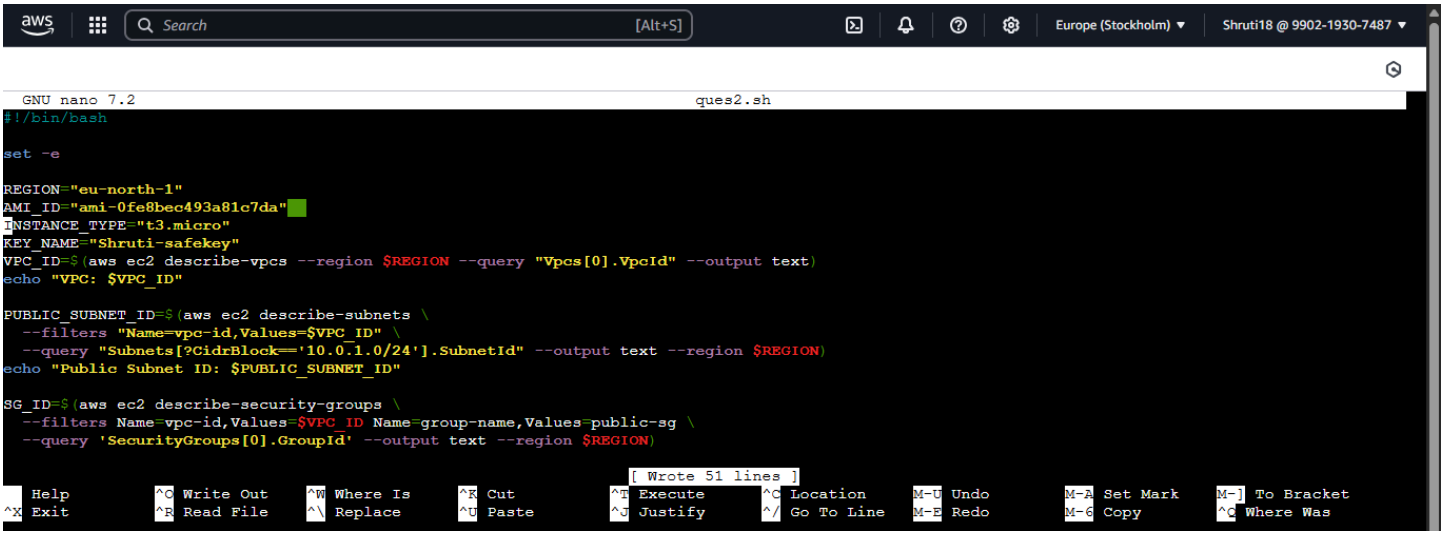
Question 02:

Write a Bash Script that fetches the id of the public subnet, create an EC2 Instance and attach this Instance to the public subnet. Create an alarm for CPU utilisation which triggers when the threshold is greater than 70%.

Creates an SNS topic and attaches the SNS topic to the alarm, finally it attaches the alarm to the Instance created to the public subnet.

```
ubuntu@ip-172-31-28-252:~$ nano ques2.sh
ubuntu@ip-172-31-28-252:~$ chmod +x ques2.sh
ubuntu@ip-172-31-28-252:~$ bash ques2.sh
VPC: vpc-071aa108f8e7c2157
Public Subnet ID: subnet-0207ec68ed88ae5a6
Launched new instance: i-00c92016fee00f61d
 Alarm created: HighCPUAlarm-i-00c92016fee00f61d
ubuntu@ip-172-31-28-252:~$
```





Question 03:
Follow the documentation and create a dashboard with more than 5 alarms for the Instance created in the previous question.

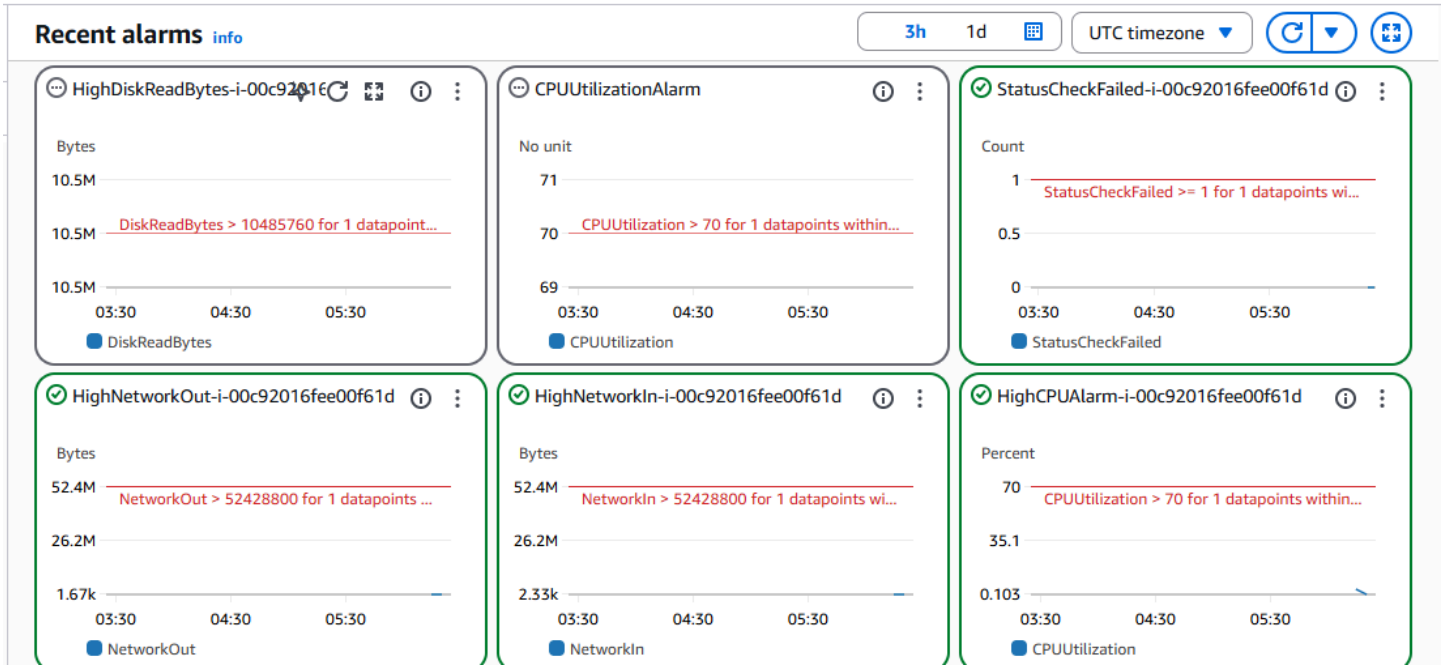
```
ubuntu@ip-172-31-28-252:~$ bash ques3.sh
Using Instance ID: i-00c92016fee00f61d
{
    "DashboardValidationMessages": []
}
 Dashboard 'EC2MonitoringDashboard' created with 5 metrics
ubuntu@ip-172-31-28-252:~$
```

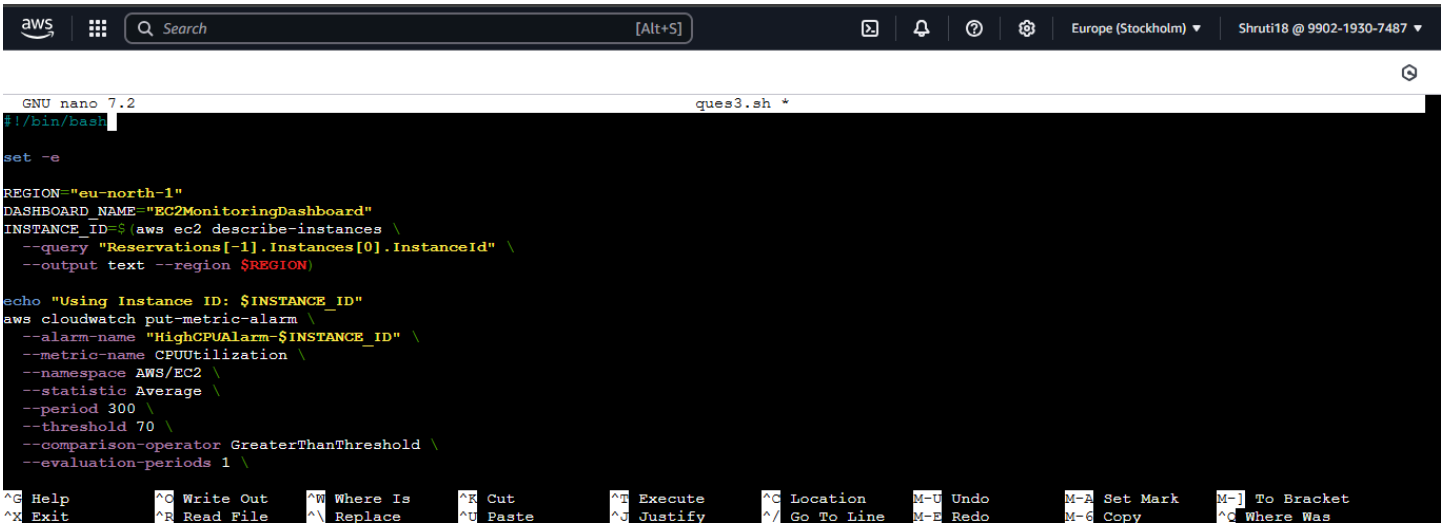
▼ Alarms

Find alarms by name

1

Name	State	Description	Metric name	State reason
HighCPUAlarm-...	OK	–	CPUUtilization	-
StatusCheckFail...	OK	–	StatusCheckFa...	-
HighNetworkIn-...	OK	–	NetworkIn	-
HighNetworkO...	OK	–	NetworkOut	-




aws
Search
[Alt+S]
Europe (Stockholm)
Shruti18 @ 9902-1930-7487
GNU nano 7.2
ques3.sh *
#!/bin/bash
set -e
REGION="eu-north-1"
DASHBOARD_NAME="EC2MonitoringDashboard"
INSTANCE_ID=$(aws ec2 describe-instances \
--query "Reservations[-1].Instances[0].InstanceId" \
--output text --region $REGION)
echo "Using Instance ID: $INSTANCE_ID"
aws cloudwatch put-metric-alarm \
--alarm-name "HighCPUAlarm-$INSTANCE_ID" \
--metric-name CPUUtilization \
--namespace AWS/EC2 \
--statistic Average \
--period 300 \
--threshold 70 \
--comparison-operator GreaterThanThreshold \
--evaluation-periods 1 \
^G Help
^O Write Out
^W Where Is
^K Cut
^T Execute
^C Location
M-U Undo
M-A Set Mark
M-] To Bracket
^X Exit
^R Read File
^\ Replace
^U Paste
^J Justify
^/ Go To Line
M-E Redo
M-6 Copy
^Q Where Was

AWS ASSIGNMENT
TOPIC:DOCKER

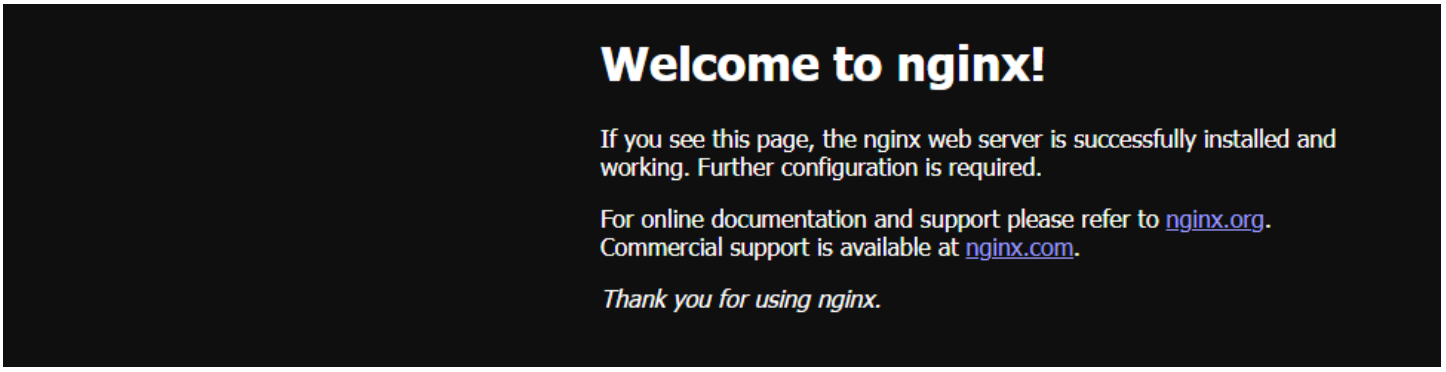
Name: Shruti Das
Registration no.: 2241004105
Date: 25/07/2025

1. Pull and run the hello-world image.

```
ubuntu@ip-172-31-33-236:~$ sudo docker pull hello-world
Using default tag: latest
latest: Pulling from library/hello-world
e6590344b1a5: Pull complete
Digest: sha256:ec153840d1e635ac434fab5e377081f17e0e15afab27beb3f726c3265039cfff
Status: Downloaded newer image for hello-world:latest
docker.io/library/hello-world:latest
ubuntu@ip-172-31-33-236:~$ sudo docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.
```

2. Pull and run the nginx image, check if the webpage is actively hosted.




```
ubuntu@ip-172-31-33-236:~$ sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
59e22667830b: Pull complete
140da4f89dcb: Pull complete
96e47e70491e: Pull complete
2ef442a3816e: Pull complete
4b1e45a9989f: Pull complete
1d9f51194194: Pull complete
f30ffbee4c54: Pull complete
Digest: sha256:84ec966e61a8c7846f509da7eb081c55c1d568174487289 24a87ab32f12a72fb
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
ubuntu@ip-172-31-33-236:~$ sudo docker run -d -p 80:80 --name mynginx nginx
6582ece01da468c541f76d3a532bfb03f23863ca07ab812bd8ba170a57893401
```

3. Pull and run mysql container from hub. Can you log into the tty provided using -it option.

```
ubuntu@ip-172-31-33-236:~$ sudo docker pull mysql
Using default tag: latest
latest: Pulling from library/mysql
3ea90c3a51a3: Pull complete
afaae28bbee0: Pull complete
2ba13c6c03c3: Pull complete
31c5b5c14efc: Pull complete
938fdecc6487: Pull complete
19f77cf53703: Pull complete
6c4bb10a796d: Pull complete
468bb678f509: Pull complete
90fd4b47993e: Pull complete
6dfd9f336bec: Pull complete
Digest: sha256:082063dca94535c76b91c6ef9b9f6748810e546887af5e8574a8cfa11eace184
Status: Downloaded newer image for mysql:latest
```

```
ubuntu@ip-172-31-33-236:~$ sudo docker run -d \
  --name mysql-container \
  -e MYSQL_ROOT_PASSWORD=my-secret-pw \
  mysql
d993e8f45b5886e632b181166d546488426278fe931ca4ab18ad29b094ce1ccc
```

```
ubuntu@ip-172-31-33-236:~$ sudo docker exec -it mysql-container mysql -uroot -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 9.4.0 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

4. Pull the nginx image from docker hub, rename the image to mynginx and upload it to docker hub.

```
ubuntu@ip-172-31-33-236:~$ sudo docker tag nginx shruti1804/mynginx
ubuntu@ip-172-31-33-236:~$ sudo docker images
REPOSITORY           TAG       IMAGE ID       CREATED        SIZE
mysql                latest    fe7f726d39a6   10 days ago    921MB
nginx                latest    2cd1d97f893f   2 weeks ago    192MB
shruti1804/mynginx   latest    2cd1d97f893f   2 weeks ago    192MB
hello-world          latest    74cc54e27dc4   6 months ago   10.1kB
ubuntu@ip-172-31-33-236:~$ sudo docker login

USING WEB-BASED LOGIN

i Info → To sign in with credentials on the command line, use 'docker login -u <username>'


Your one-time device confirmation code is: JKQL-ZCTW
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser…

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
```

```
ubuntu@ip-172-31-33-236:~$ sudo docker push shruti1804/mynginx
Using default tag: latest
The push refers to repository [docker.io/shruti1804/mynginx]
f17478b6e8f3: Mounted from library/nginx
0662742b23b2: Mounted from library/nginx
5c91a024d899: Mounted from library/nginx
6b1b97dc9285: Mounted from library/nginx
a6b19c3d00b1: Mounted from library/nginx
30837a0774b9: Mounted from library/nginx
7cc7fe68eff6: Mounted from library/nginx
latest: digest: sha256:c6185f1da234e4b891492e4c60e5b06093d327ed19a1b90318f4882160ffd0bd size: 1778
ubuntu@ip-172-31-33-236:~$
```

5.Create a EC2 instance and then install docker in it. Run the getting started app and create a container out of. Upload your container image to docker hub. Can you run the container created by you in any other system that runs docker engine.

6.Write a bash script to create VPC and a subnet, fulfil all network requirements (route tables, routes and IG). Pass another script to the instance that installs and configures docker and install msql. Create and display a database in it.

```
ubuntu@ip-172-31-33-236:~$ bash create_infra.sh
✅VPC created: vpc-0d54f9c3d6666965d
✅Subnet created: subnet-078045db4e9fd7405
✅IGW created and attached: igw-0537740f20f2edec0
{
    "Return": true
}
{
    "AssociationId": "rtbassoc-007fa1e3efc2569bd",
    "AssociationState": {
        "State": "associated"
    }
}
✅Route table created and associated: rtb-0678dab07624bece5
{
    "Return": true,
    "SecurityGroupRules": [
        {
            "SecurityGroupRuleId": "sgr-0e9a005ef1add90cb",
            "GroupId": "sg-0ee3475082743f5f3",
            "GroupOwnerId": "990219307487",
            "IsEgress": false,
            "IpProtocol": "tcp",
            "FromPort": 22,
            "CidrIpv4": "0.0.0.0/0",
            "SecurityGroupRuleArn": "arn:aws:ec2:eu-north-1:990219307487:security-group-rule/sgr-0e9a005ef1add90cb"
        }
    ]
}
{
    "Return": true,
    "SecurityGroupRules": [
        {
            "SecurityGroupRuleId": "sgr-0d7fc1690fce0d0ac",
            "GroupId": "sg-0ee3475082743f5f3",
            "GroupOwnerId": "990219307487",
            "IsEgress": false,
            "IpProtocol": "tcp",
            "FromPort": 3306,
            "ToPort": 3306,
            "CidrIpv4": "0.0.0.0/0",
            "SecurityGroupRuleArn": "arn:aws:ec2:eu-north-1:990219307487:security-group-rule/sgr-0d7fc1690fce0d0ac"
        }
    ]
}
✅Security Group created: sg-0ee3475082743f5f3
```

```
# 1. Create VPC
VPC_ID=$(aws ec2 create-vpc --cidr-block $VPC_CIDR \
  --query 'Vpc.VpcId' --output text --region $REGION)

aws ec2 modify-vpc-attribute --vpc-id $VPC_ID --enable-dns-support "{\"Value\":true}" --region $REGION
aws ec2 modify-vpc-attribute --vpc-id $VPC_ID --enable-dns-hostnames "{\"Value\":true}" --region $REGION

echo "✅VPC created: $VPC_ID"

# 2. Create Subnet
SUBNET_ID=$(aws ec2 create-subnet --vpc-id $VPC_ID \
  --cidr-block $SUBNET_CIDR --availability-zone ${REGION}a \
  --query 'Subnet.SubnetId' --output text --region $REGION)

echo "✅Subnet created: $SUBNET_ID"

# 3. Create Internet Gateway and Attach
IGW_ID=$(aws ec2 create-internet-gateway \
  --query 'InternetGateway.InternetGatewayId' --output text --region $REGION)
aws ec2 attach-internet-gateway --internet-gateway-id $IGW_ID --vpc-id $VPC_ID --region $REGION

echo "✅IGW created and attached: $IGW_ID"

# 4. Create Route Table and associate
ROUTE_TABLE_ID=$(aws ec2 create-route-table --vpc-id $VPC_ID \
  --query 'RouteTable.RouteTableId' --output text --region $REGION)

aws ec2 create-route --route-table-id $ROUTE_TABLE_ID \
  --destination-cidr-block 0.0.0.0/0 --gateway-id $IGW_ID --region $REGION
```

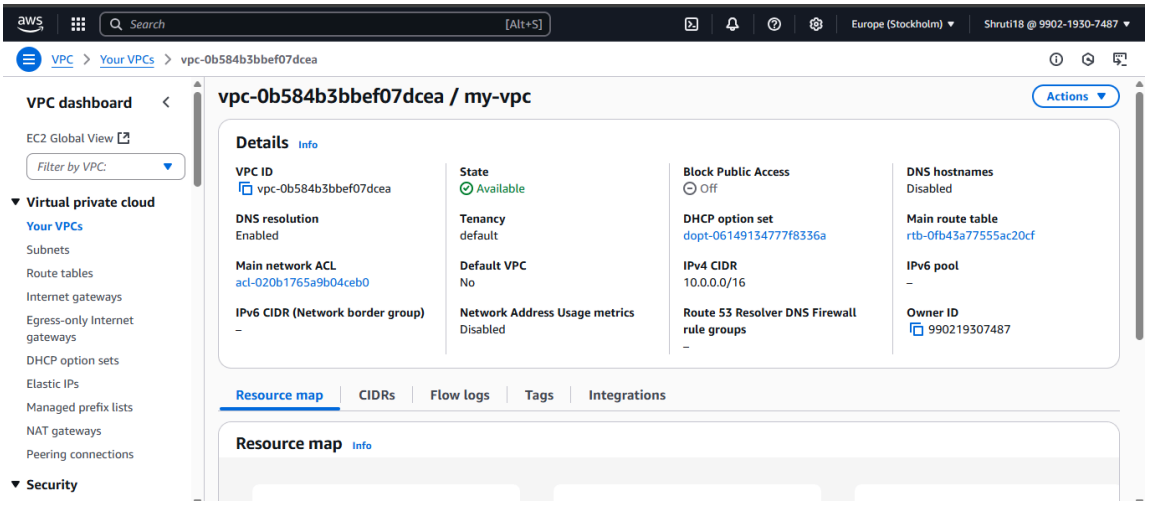
```
ubuntu@ip-172-31-33-236:~$ cat install_mysql.sh
#!/bin/bash
apt update && apt install -y docker.io

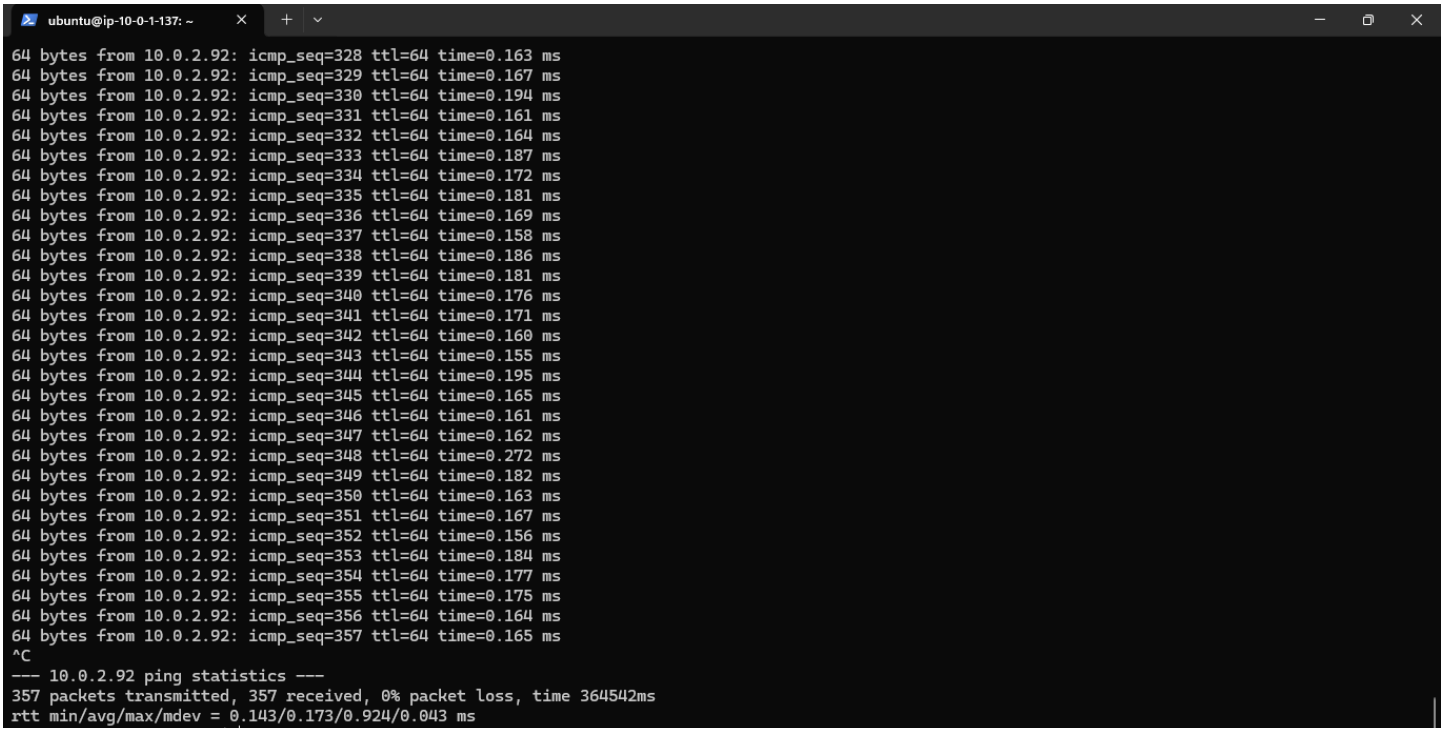
docker run -d \
  --name mysql-server \
  -e MYSQL_ROOT_PASSWORD=rootpass \
  -e MYSQL_DATABASE=devopsdb \
  mysql

# Wait a bit to let MySQL initialize
sleep 30

docker exec -i mysql-server \
  mysql -uroot -prootpass -e "SHOW DATABASES;" > /home/ubuntu/db_list.txt
```

10.Create 2 instances in two different subnets but within the same VPC, show that instances within a VPC can communicate with each other(ssh, ping).

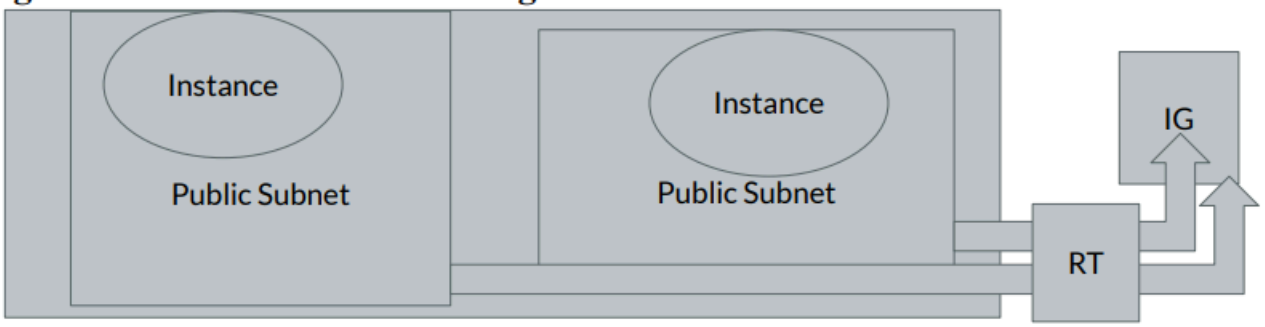


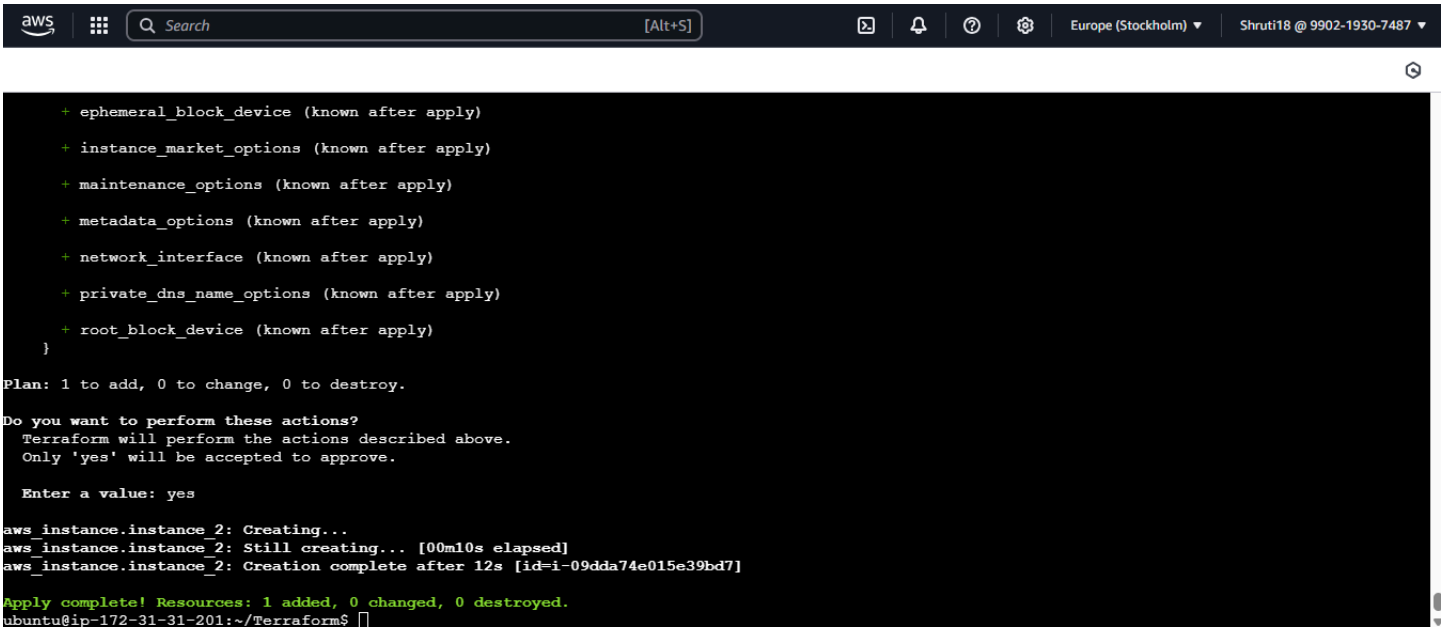

ubuntu@ip-10-0-1-137: ~
64 bytes from 10.0.2.92: icmp_seq=328 ttl=64 time=0.163 ms
64 bytes from 10.0.2.92: icmp_seq=329 ttl=64 time=0.167 ms
64 bytes from 10.0.2.92: icmp_seq=330 ttl=64 time=0.194 ms
64 bytes from 10.0.2.92: icmp_seq=331 ttl=64 time=0.161 ms
64 bytes from 10.0.2.92: icmp_seq=332 ttl=64 time=0.164 ms
64 bytes from 10.0.2.92: icmp_seq=333 ttl=64 time=0.187 ms
64 bytes from 10.0.2.92: icmp_seq=334 ttl=64 time=0.172 ms
64 bytes from 10.0.2.92: icmp_seq=335 ttl=64 time=0.181 ms
64 bytes from 10.0.2.92: icmp_seq=336 ttl=64 time=0.169 ms
64 bytes from 10.0.2.92: icmp_seq=337 ttl=64 time=0.158 ms
64 bytes from 10.0.2.92: icmp_seq=338 ttl=64 time=0.186 ms
64 bytes from 10.0.2.92: icmp_seq=339 ttl=64 time=0.181 ms
64 bytes from 10.0.2.92: icmp_seq=340 ttl=64 time=0.176 ms
64 bytes from 10.0.2.92: icmp_seq=341 ttl=64 time=0.171 ms
64 bytes from 10.0.2.92: icmp_seq=342 ttl=64 time=0.160 ms
64 bytes from 10.0.2.92: icmp_seq=343 ttl=64 time=0.155 ms
64 bytes from 10.0.2.92: icmp_seq=344 ttl=64 time=0.195 ms
64 bytes from 10.0.2.92: icmp_seq=345 ttl=64 time=0.165 ms
64 bytes from 10.0.2.92: icmp_seq=346 ttl=64 time=0.161 ms
64 bytes from 10.0.2.92: icmp_seq=347 ttl=64 time=0.162 ms
64 bytes from 10.0.2.92: icmp_seq=348 ttl=64 time=0.272 ms
64 bytes from 10.0.2.92: icmp_seq=349 ttl=64 time=0.182 ms
64 bytes from 10.0.2.92: icmp_seq=350 ttl=64 time=0.163 ms
64 bytes from 10.0.2.92: icmp_seq=351 ttl=64 time=0.167 ms
64 bytes from 10.0.2.92: icmp_seq=352 ttl=64 time=0.156 ms
64 bytes from 10.0.2.92: icmp_seq=353 ttl=64 time=0.184 ms
64 bytes from 10.0.2.92: icmp_seq=354 ttl=64 time=0.177 ms
64 bytes from 10.0.2.92: icmp_seq=355 ttl=64 time=0.175 ms
64 bytes from 10.0.2.92: icmp_seq=356 ttl=64 time=0.164 ms
64 bytes from 10.0.2.92: icmp_seq=357 ttl=64 time=0.165 ms
^C
--- 10.0.2.92 ping statistics ---
357 packets transmitted, 357 received, 0% packet loss, time 364542ms
rtt min/avg/max/mdev = 0.143/0.173/0.924/0.043 ms

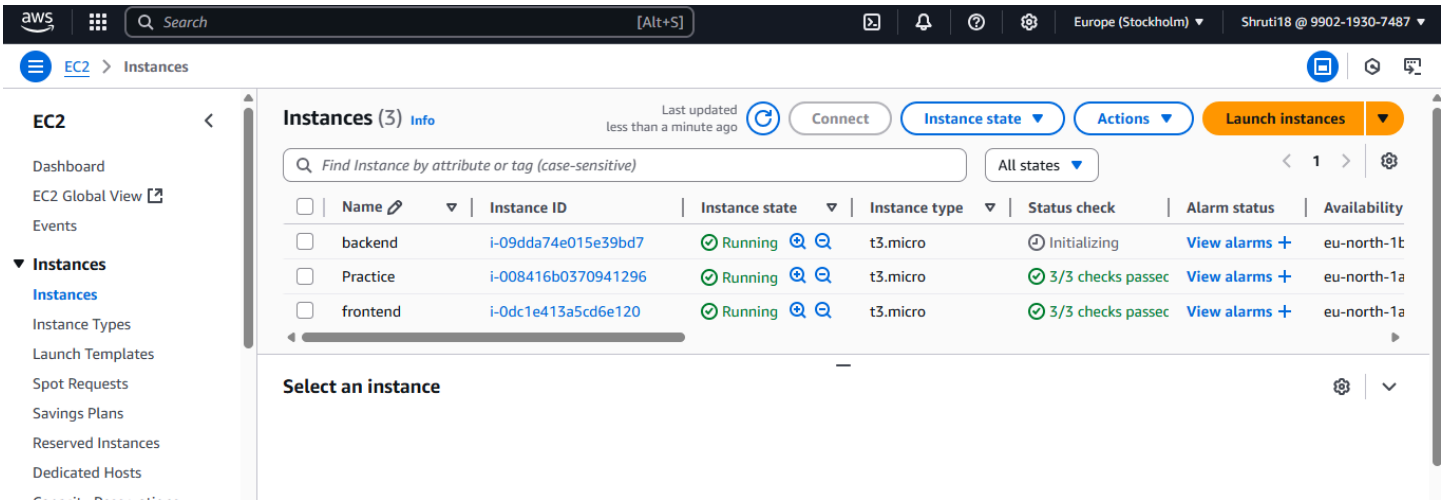
AWS ASSIGNMENT
TOPIC:TERRAFORM

Name: Shruti Das
Registration no.: 2241004105
Date: 25/07/2025

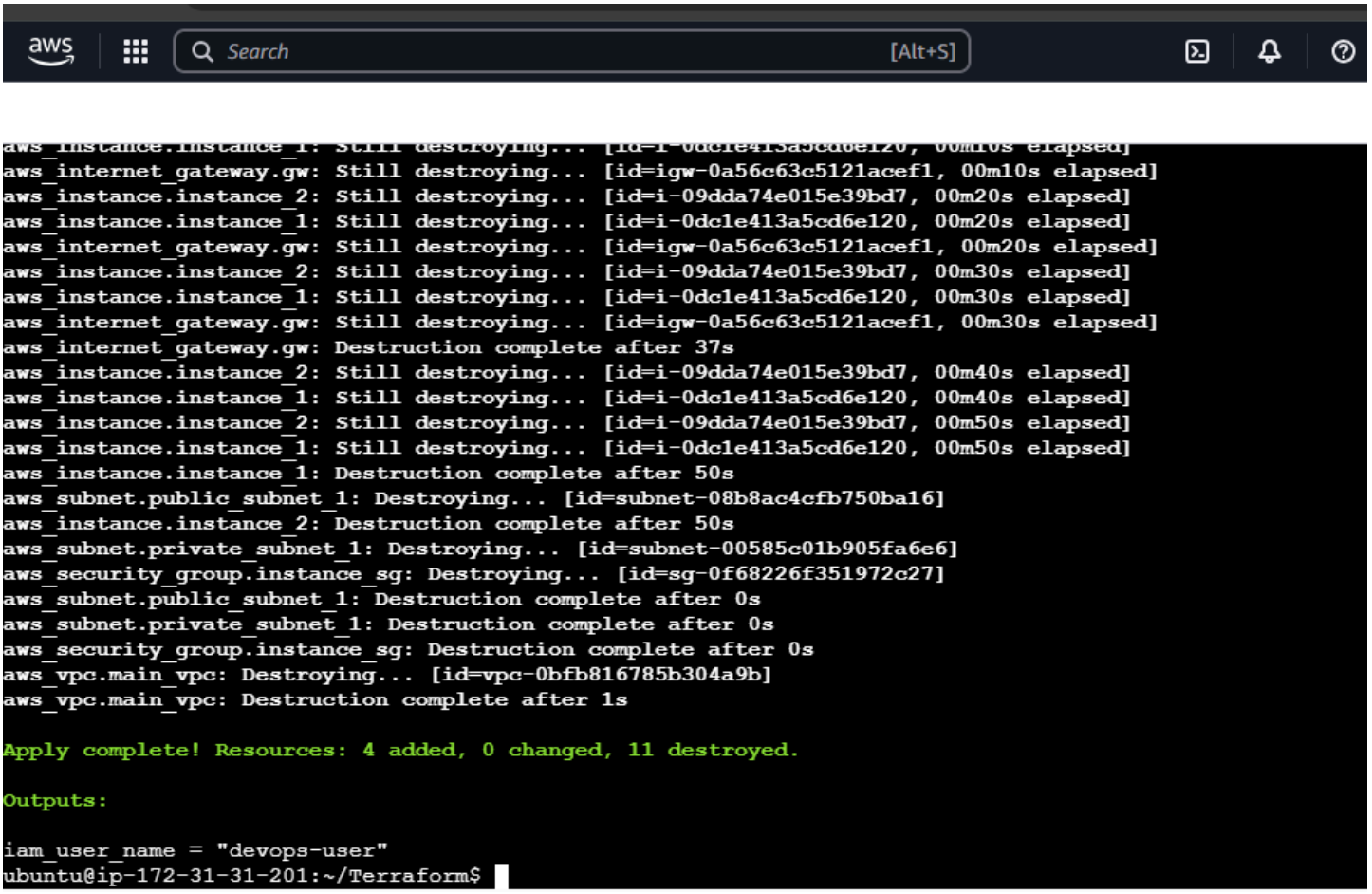
1. Using terraform create the following architecture

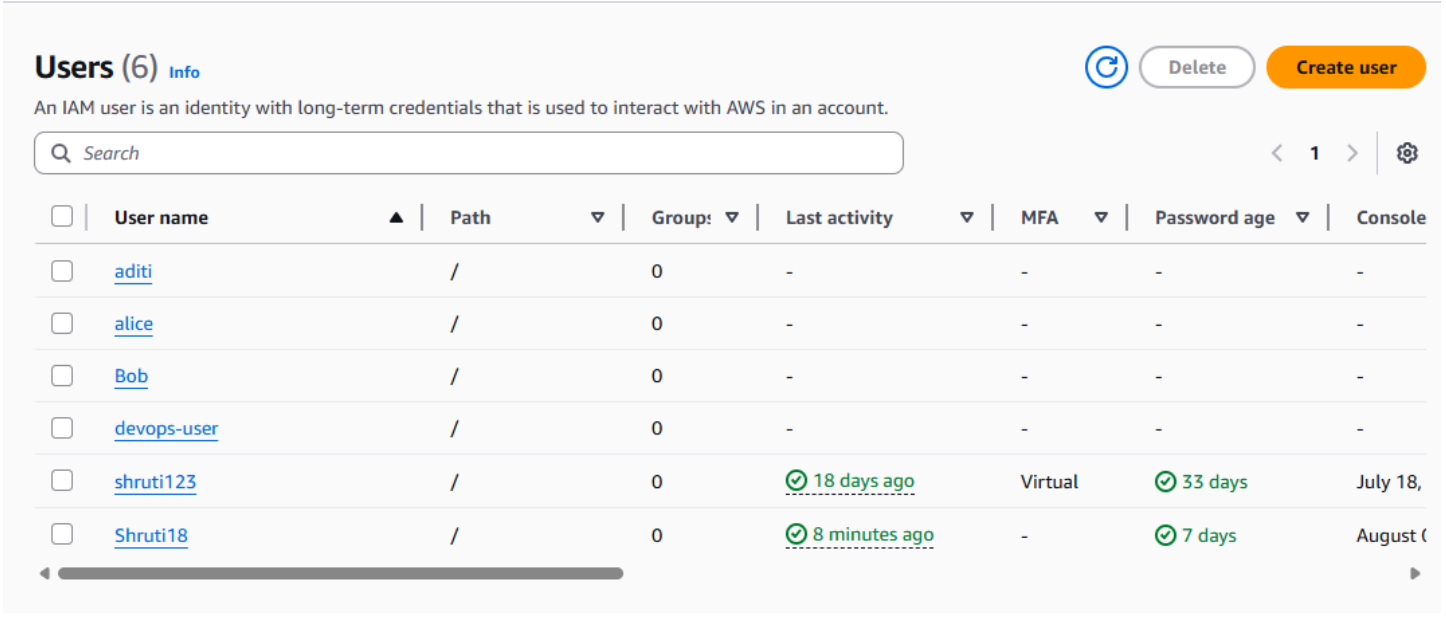






2. Using terraform create an IAM user and provide the following permissions:
a. EC2fullaccess
b. Administratoraccess
c. S3fullaccess


aws
Search
[Alt+S]
aws_instance.instance_1: Still destroying... [id=i-0dc1e413a5cd6e120, 00m10s elapsed]
aws_internet_gateway.gw: Still destroying... [id=igw-0a56c63c5121acef1, 00m10s elapsed]
aws_instance.instance_2: Still destroying... [id=i-09dda74e015e39bd7, 00m20s elapsed]
aws_instance.instance_1: Still destroying... [id=i-0dc1e413a5cd6e120, 00m20s elapsed]
aws_internet_gateway.gw: Still destroying... [id=igw-0a56c63c5121acef1, 00m20s elapsed]
aws_instance.instance_2: Still destroying... [id=i-09dda74e015e39bd7, 00m30s elapsed]
aws_instance.instance_1: Still destroying... [id=i-0dc1e413a5cd6e120, 00m30s elapsed]
aws_internet_gateway.gw: Still destroying... [id=igw-0a56c63c5121acef1, 00m30s elapsed]
aws_internet_gateway.gw: Destruction complete after 37s
aws_instance.instance_2: Still destroying... [id=i-09dda74e015e39bd7, 00m40s elapsed]
aws_instance.instance_1: Still destroying... [id=i-0dc1e413a5cd6e120, 00m40s elapsed]
aws_instance.instance_2: Still destroying... [id=i-09dda74e015e39bd7, 00m50s elapsed]
aws_instance.instance_1: Still destroying... [id=i-0dc1e413a5cd6e120, 00m50s elapsed]
aws_instance.instance_1: Destruction complete after 50s
aws_subnet.public_subnet_1: Destroying... [id=subnet-08b8ac4cfb750ba16]
aws_instance.instance_2: Destruction complete after 50s
aws_subnet.private_subnet_1: Destroying... [id=subnet-00585c01b905fa6e6]
aws_security_group.instance_sg: Destroying... [id=sg-0f68226f351972c27]
aws_subnet.public_subnet_1: Destruction complete after 0s
aws_subnet.private_subnet_1: Destruction complete after 0s
aws_security_group.instance_sg: Destruction complete after 0s
aws_vpc.main_vpc: Destroying... [id=vpc-0bfb816785b304a9b]
aws_vpc.main_vpc: Destruction complete after 1s
Apply complete! Resources: 4 added, 0 changed, 11 destroyed.
Outputs:
iam_user_name = "devops-user"
ubuntu@ip-172-31-31-201:~/Terraform$


Users (6) Info
An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.
Delete
Create user
Search
1
User name
Path
Group:
Last activity
MFA
Password age
Console
aditi / 0 - - - -
alice / 0 - - - -
Bob / 0 - - - -
devops-user / 0 - - - -
shruti123 / 0 18 days ago Virtual 33 days July 18,
Shruti18 / 0 8 minutes ago - 7 days August (

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DEEMED TO BE UNIVERSITY

DevOps Summer Training Program 2025

Deploying a 2-tier Application on AWS Cloud Using Terraform and Jenkins

Submitted by-

Name: Shruti Das

Registration No.: 2241004105

Branch: B Tech CSE

Admission Batch: 2022

Session: 2025-2026

Serial No.: 47

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1. Introduction

In modern software development, DevOps practices are widely adopted to improve deployment speed, maintain application stability, and reduce manual errors.

The app is a 2-tier web application, which means it has a frontend (what the user sees) and a backend (where the data is stored and managed). The goal is to deploy both parts on AWS EC2 instances — the frontend in a public subnet and the backend in a private subnet.

I divided the whole project into 3 stages:

1. **Create_Infra** : setting up the cloud infrastructure.
2. **Deploy_Apps** : running the application using Docker.
3. **Test_Solution** : checking if everything works properly.

Each part was automated to reduce manual effort and to make sure the setup is fast, repeatable, and reliable.

2. Objective

To automate the complete lifecycle of deploying a 2-tier application (frontend and backend/database) on AWS EC2 instances using Terraform and Jenkins, ensuring a fully containerized and tested environment with minimal manual effort.

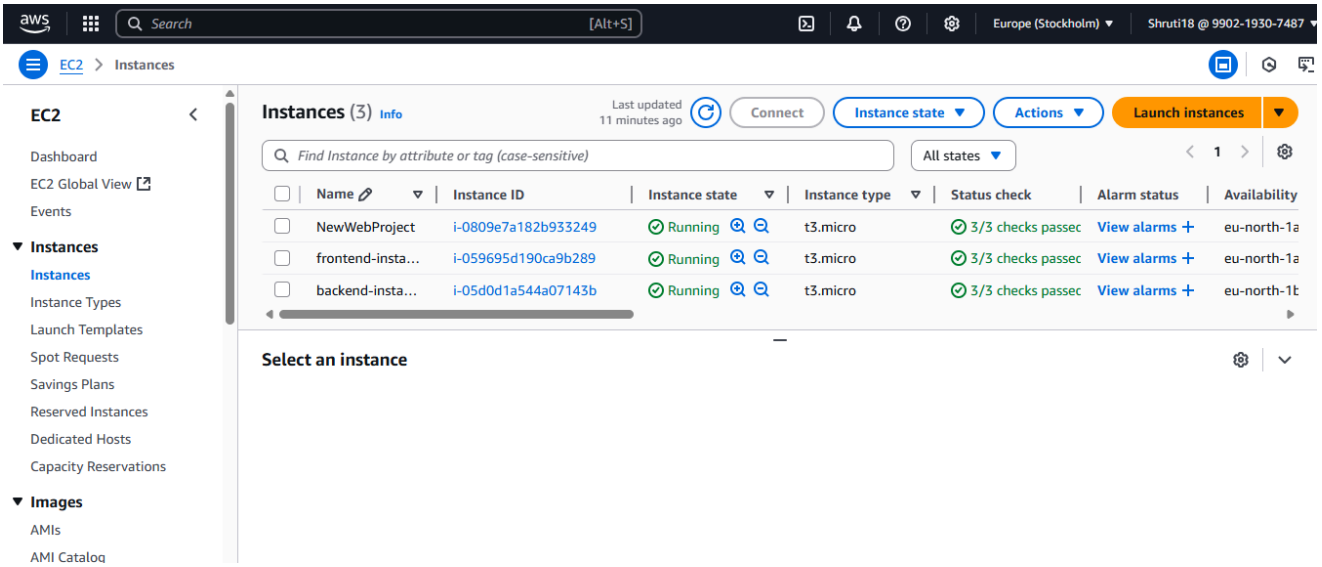
3. Tools & Technologies Used

- Terraform – For infrastructure as code (IaC) to create AWS VPC, subnets, EC2 instances, etc.
- Docker – To containerize the frontend and backend components.
- AWS EC2, VPC, IGW, Route Tables – For cloud infrastructure.
- Bash Scripting – For automating instance setup via provisioners.
- Remote-Exec Provisioner – For post-deployment checks and setup.

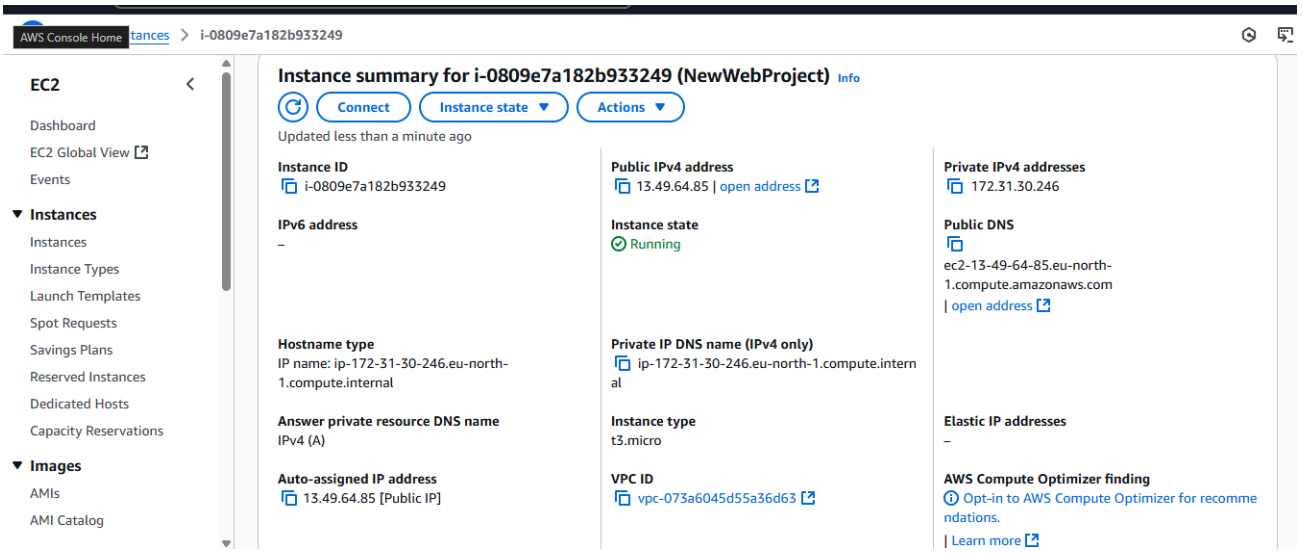
4. Project Architecture

- A Public VPC containing two subnets:
    - PUBLIC Subnet: Hosts the frontend EC2 instance.
    - PRIVATE Subnet: Hosts the backend EC2 instance (secured, no internet access).
- Communication allowed between frontend and backend within the same VPC.
- Dockerized frontend (e.g., React/HTML) interacts with backend (Node.js + MongoDB).
- Terraform scripts manage the provisioning and configuration of infrastructure.
- Jenkins automates the pipeline stages with defined triggers and steps.

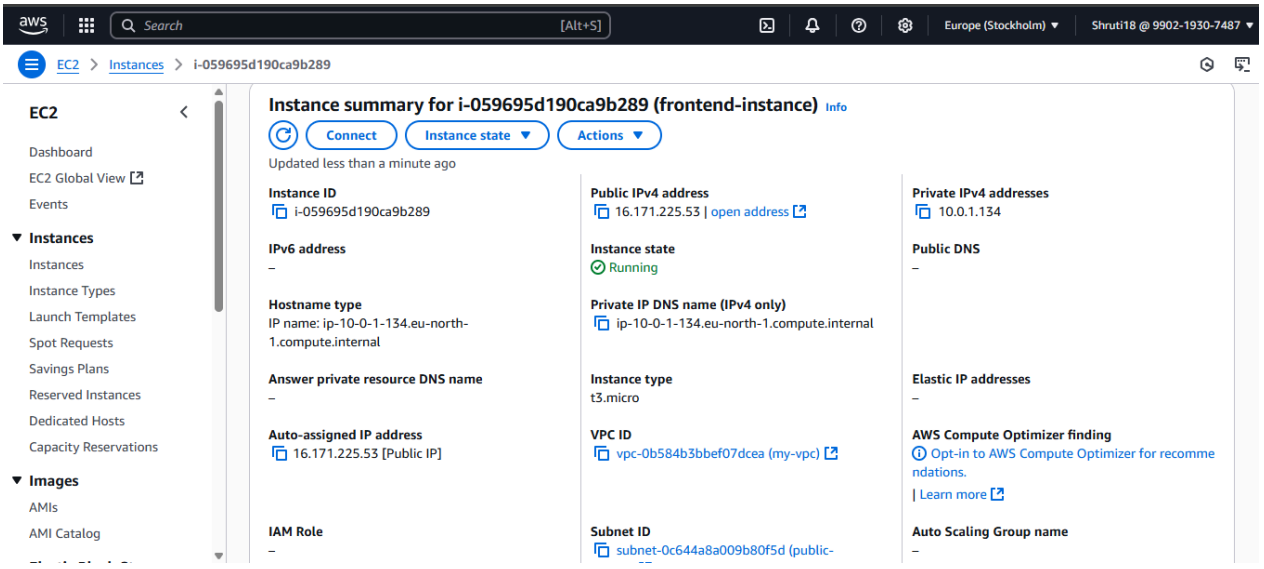
Instances:



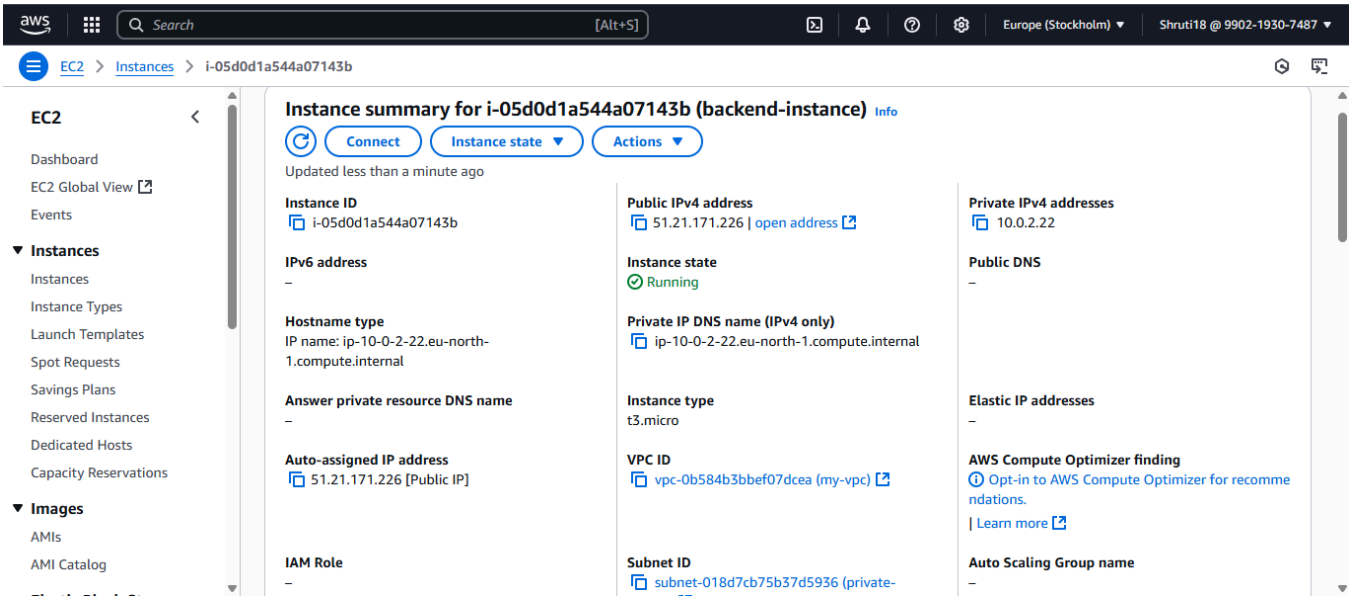
Instance: New Web Project



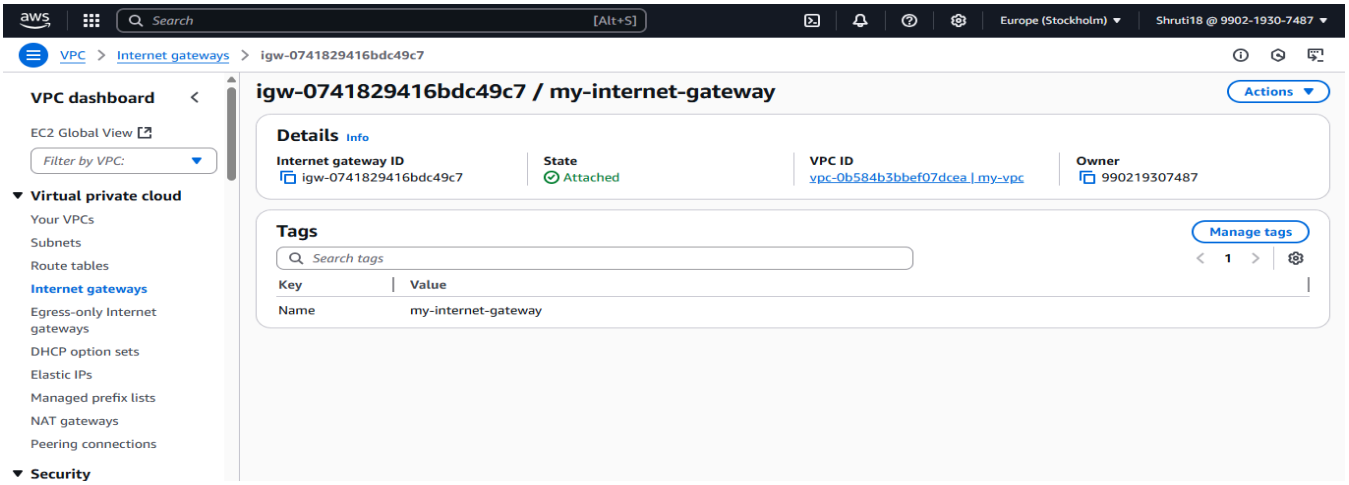
Instance: Frontend instance



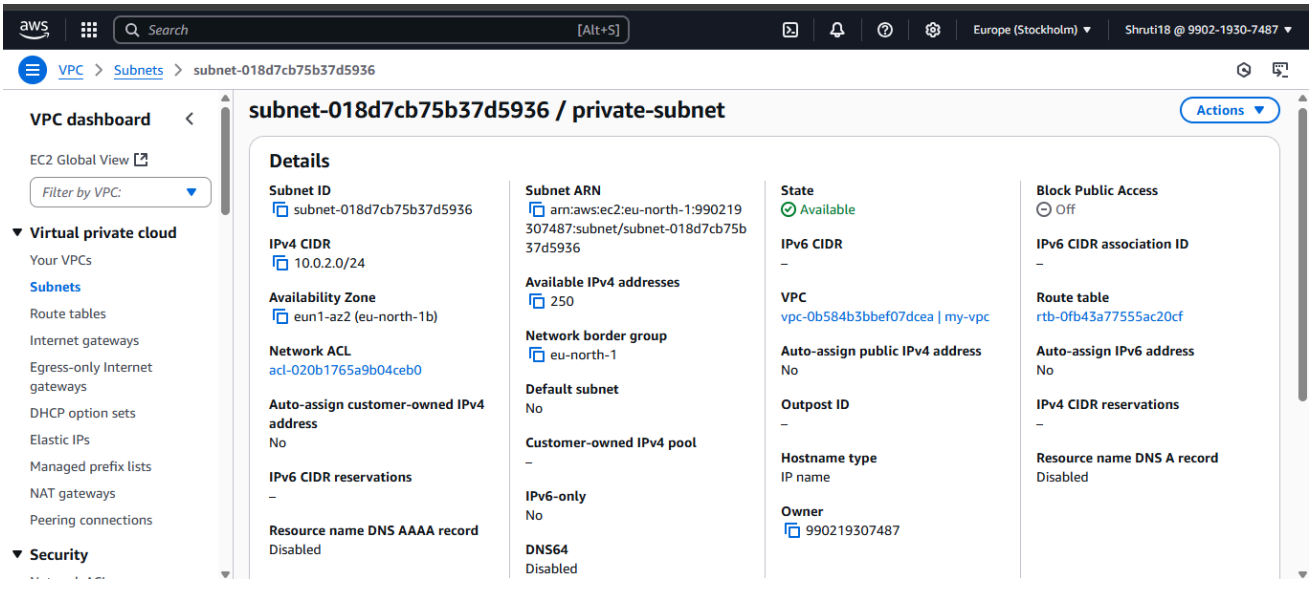
Instance: Backend Instance



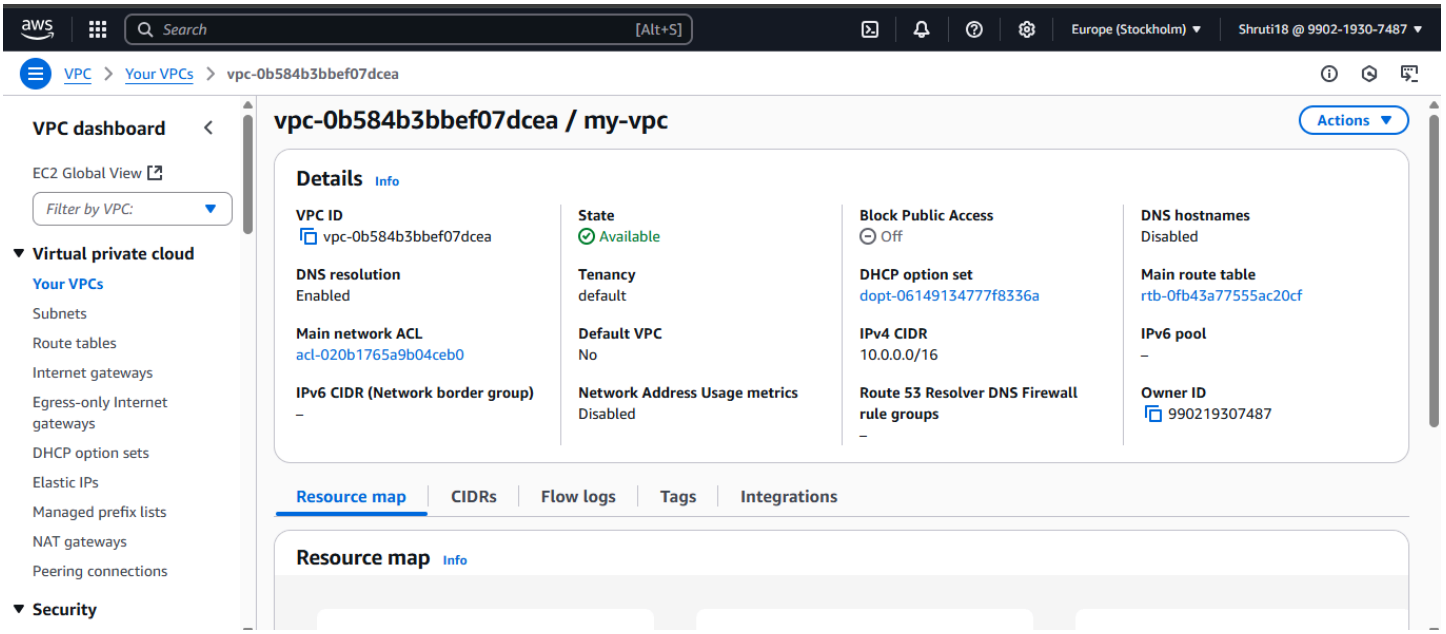
Internet Gateway:



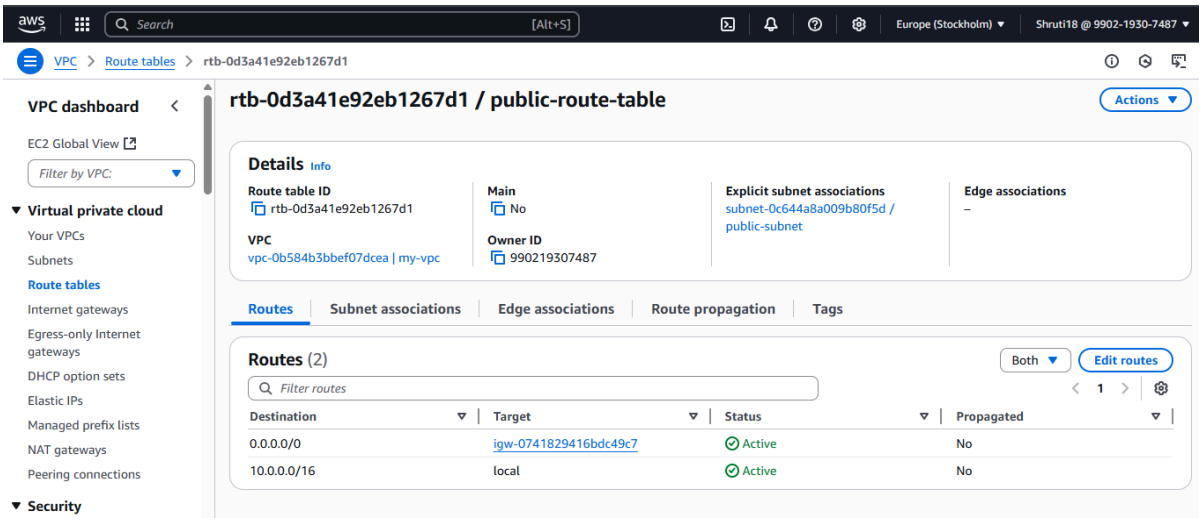
Subnets:



VPC:



Route table:



SSH:

```
PS C:\Users\shrut> ssh -i "C:\Users\shrut\Downloads\my-devops-key.pem" ubuntu@16.171.225.53
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.8.0-1029-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Jul 31 18:26:30 UTC 2025

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   33.3% of 6.71GB   Processes:             109
  Memory usage: 28%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 10.0.1.134

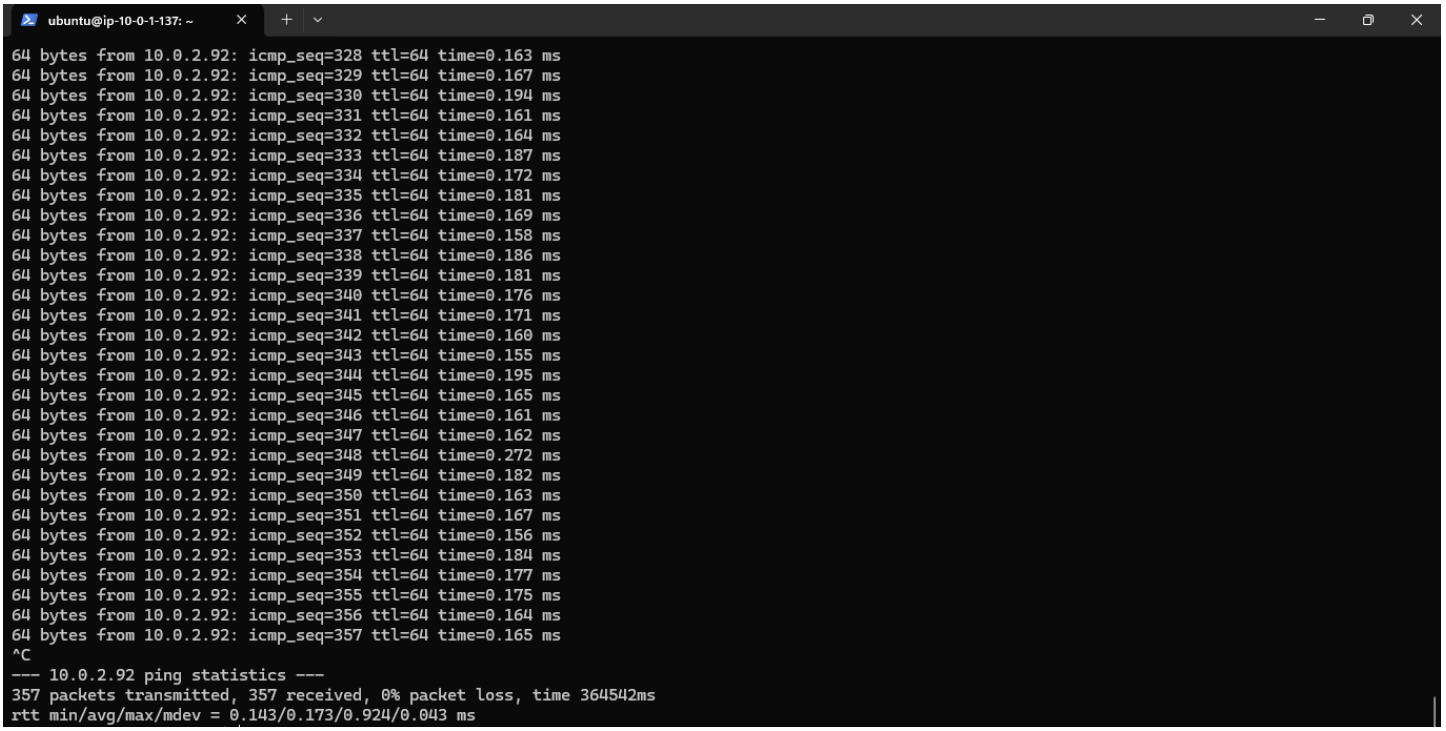

Expanded Security Maintenance for Applications is not enabled.

103 updates can be applied immediately.
66 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

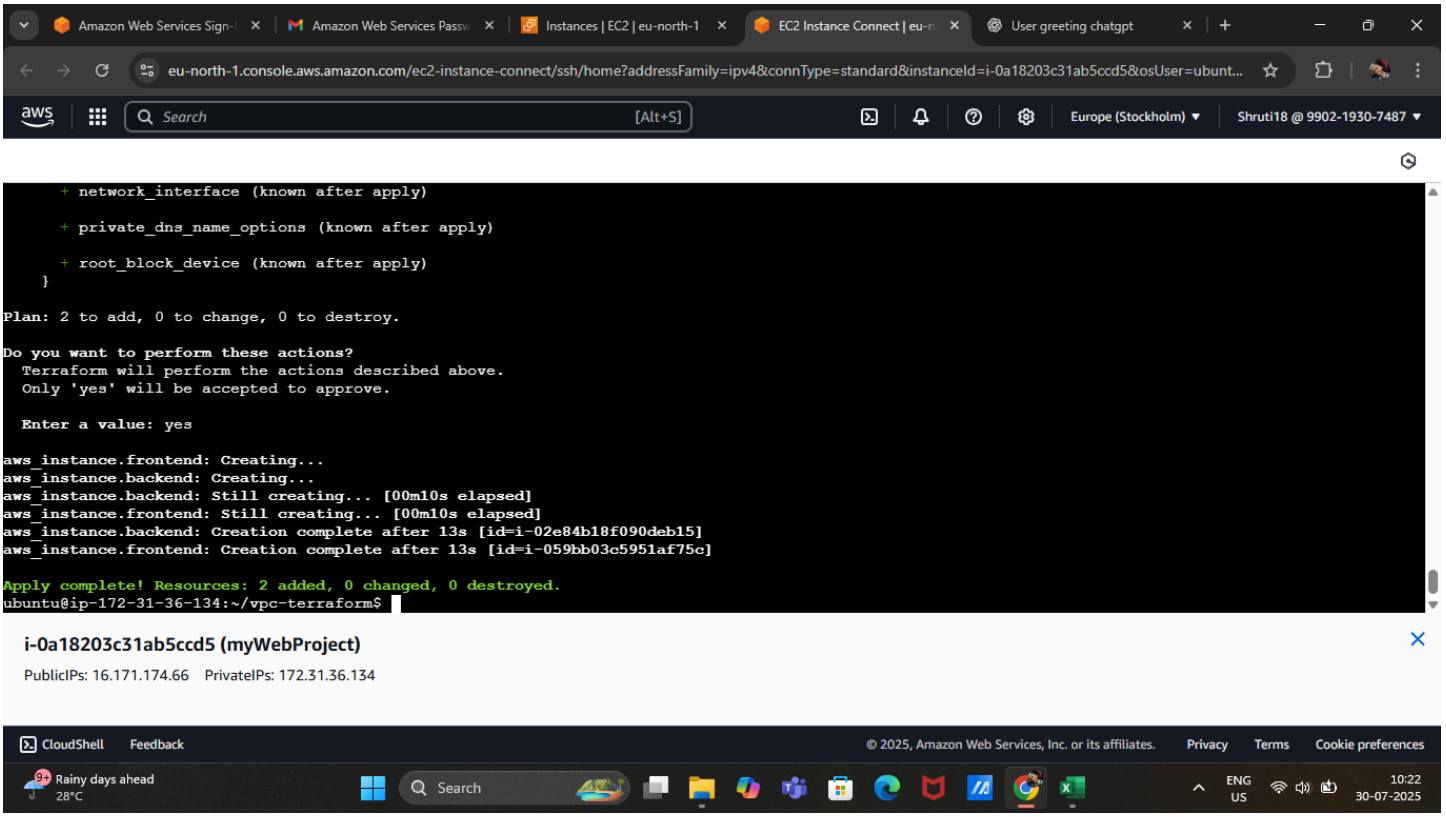
1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm


Last login: Thu Jul 31 17:44:22 2025 from 13.49.64.85
ubuntu@ip-10-0-1-134:~$
```

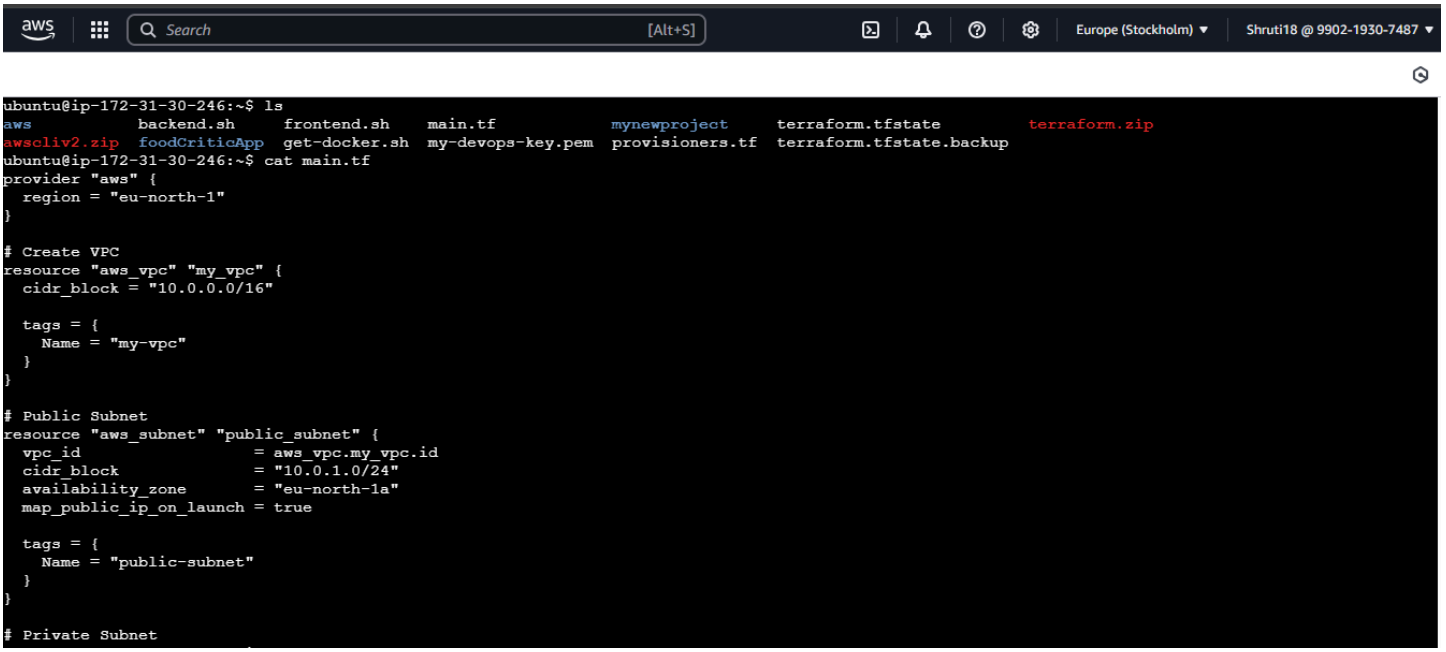
PING:



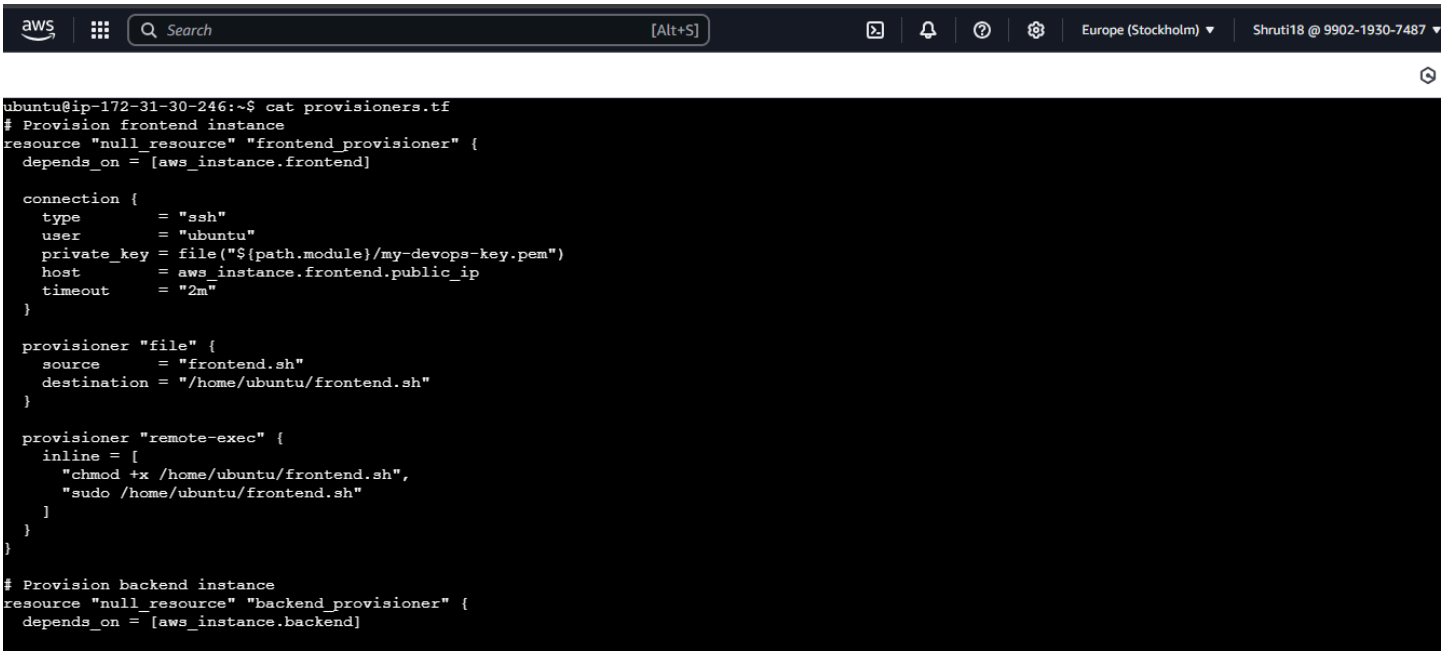
Terraform init+plan+apply



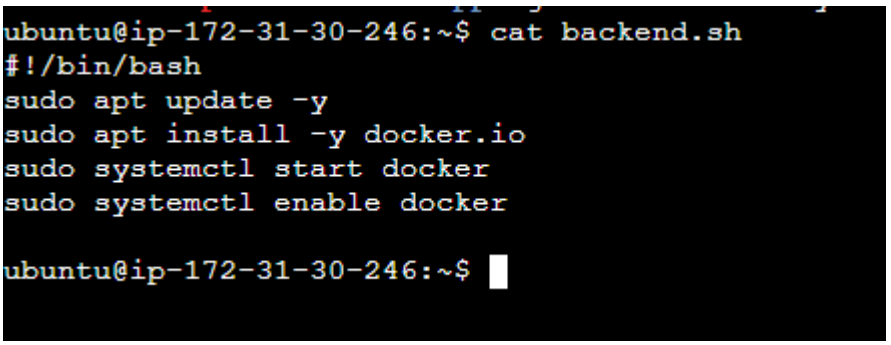
[main.tf]



provisioners.tf:



backend.sh



frontend.sh

```
ubuntu@ip-172-31-30-246:~$ cat frontend.sh
#!/bin/bash
sudo apt update -y
sudo apt install -y docker.io
sudo systemctl start docker
sudo systemctl enable docker


ubuntu@ip-172-31-30-246:~$
```

Terraform outputs:

```
}
output "frontend_public_ip" {
  value = aws_instance.frontend.public_ip
}

output "backend_private_ip" {
  value = aws_instance.backend.private_ip
}
```

Manual Testing Screenshot:

Backend:

```
^Cubuntu@ip-172-31-30-246:~/foodCriticApp/backend$ curl http://localhost:5000~
curl: (3) URL rejected: Port number was not a decimal number between 0 and 65535
ubuntu@ip-172-31-30-246:~/foodCriticApp/backend$ curl http://localhost:5000
API is running...ubuntu@ip-172-31-30-246:~/foodCriticApp/backend$
```

Frontend:

```
ubuntu@ip-172-31-30-246:~/foodCriticApp/backend$ curl http://localhost:5000
API is running...ubuntu@ip-172-31-30-246:~/foodCriticApp/backend$ ^C
ubuntu@ip-172-31-30-246:~/foodCriticApp/backend$ cd ..
ubuntu@ip-172-31-30-246:~/foodCriticApp$ cd frontend
ubuntu@ip-172-31-30-246:~/foodCriticApp/frontend$ sudo docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED          STATUS          PORTS                                       NAMES
5ddaa0a515ac   foodcritic-backend   "docker-entrypoint.s…"   16 minutes ago   Up 16 minutes   0.0.0.0:5000->3000/tcp, [::]:5000->3000/tcp   backend
22bf3af38c0a   foodcritic-frontend  "docker-entrypoint.s…"   33 minutes ago   Up 32 minutes   0.0.0.0:80->80/tcp, [::]:80->80/tcp           frontend
ubuntu@ip-172-31-30-246:~/foodCriticApp/frontend$ curl http://localhost
<!doctype html><html lang="en"><head><meta charset="utf-8"/><link rel="icon" href="https://www.shutterstock.com/image-vector/cutlery-icon-spoon-forks-
knife-600nw-1931212061.jpg"/><meta name="viewport" content="width=device-width,initial-scale=1"/><meta name="theme-color" content="#000000"/><meta nam
e="description" content="Web site created using create-react-app"/><link rel="apple-touch-icon" href="https://www.shutterstock.com/image-vector/cutler
y-icon-spoon-forks-knife-600nw-1931212061.jpg"/><link rel="manifest" href="/manifest.json"/><title>FoodCritic-SP</title><script defer="defer" src="/st
atic/js/main.48ca92d7.js"></script><link href="/static/css/main.7e487812.css" rel="stylesheet"></head><body><noscript>You need to enable JavaScript to
 run this app.</noscript><div id="root"></div></body></html>ubuntu@ip-172-31-30-246:~/foodCriticApp/frontend$
```

Testing Via Test.sh

```
ubuntu@ip-172-31-30-246:~/foodCriticApp$ nano test.sh
ubuntu@ip-172-31-30-246:~/foodCriticApp$ bash test.sh
Testing backend...
API is running...
Testing frontend...
<title>Food Critic</title>
Checking if frontend can reach backend...

[
  {
    _id: 6826412db548f75dbf3de8ef,
    name: lightnen
  }
]

ubuntu@ip-172-31-30-246:~/foodCriticApp$
```

Bash Script for Test.sh

```
ubuntu@ip-172-31-30-246:~/foodCriticApp$ cat test.sh
FRONTEND_URL="http://16.171.225.53"
BACKEND_URL="http://10.0.2.22:3000"

echo "🔍Testing backend..."
curl -s $BACKEND_URL || echo "❌Backend not reachable"

echo -e "\n🌐Testing frontend..."
curl -s $FRONTEND_URL | grep "<title>" || echo "❌Frontend not loading"

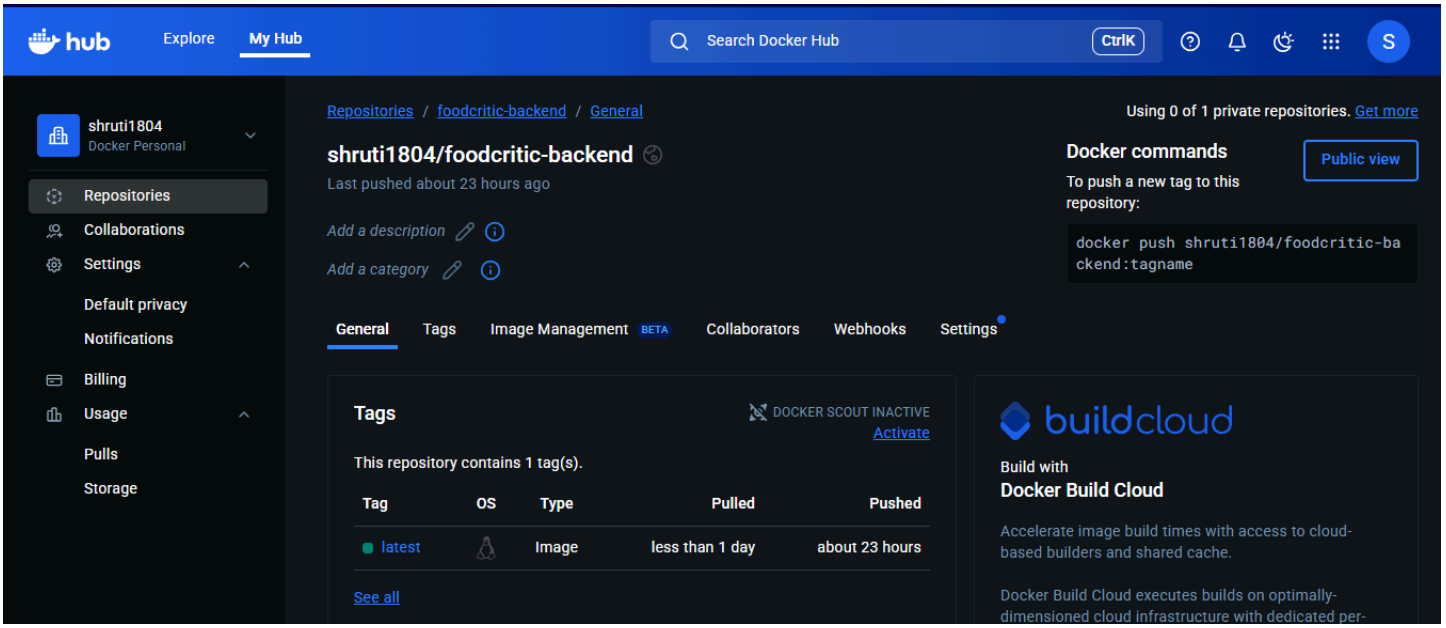
echo -e "\n🔗Checking if frontend can reach backend..."
curl -s $FRONTEND_URL/api/users || echo "❌API endpoint test failed"

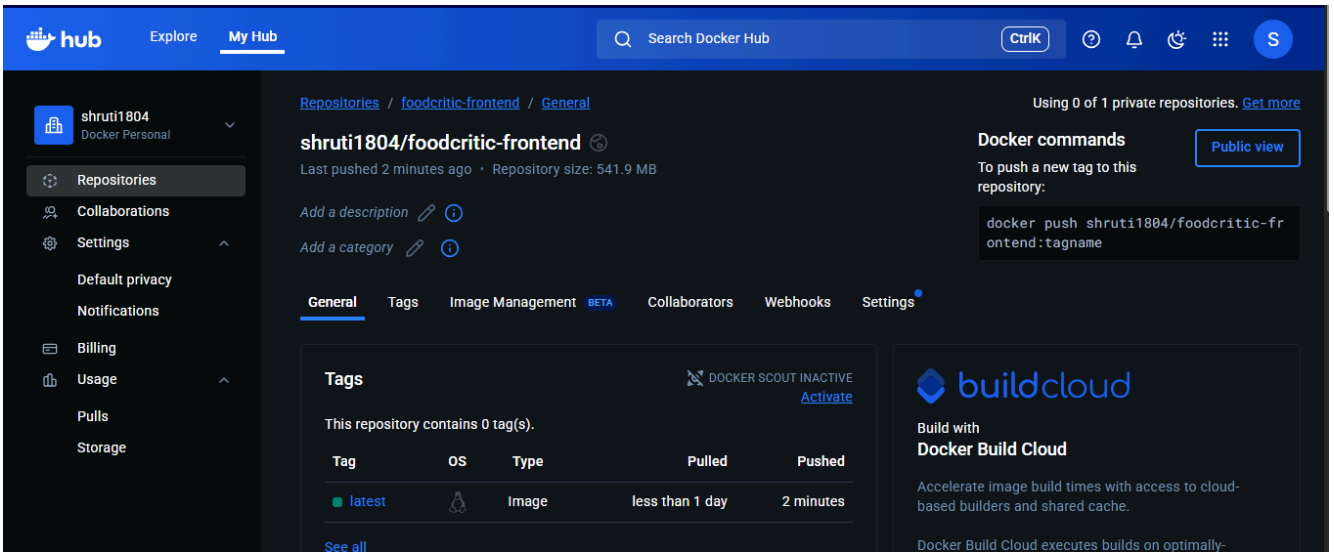
ubuntu@ip-172-31-30-246:~/foodCriticApp$
```

Docker:

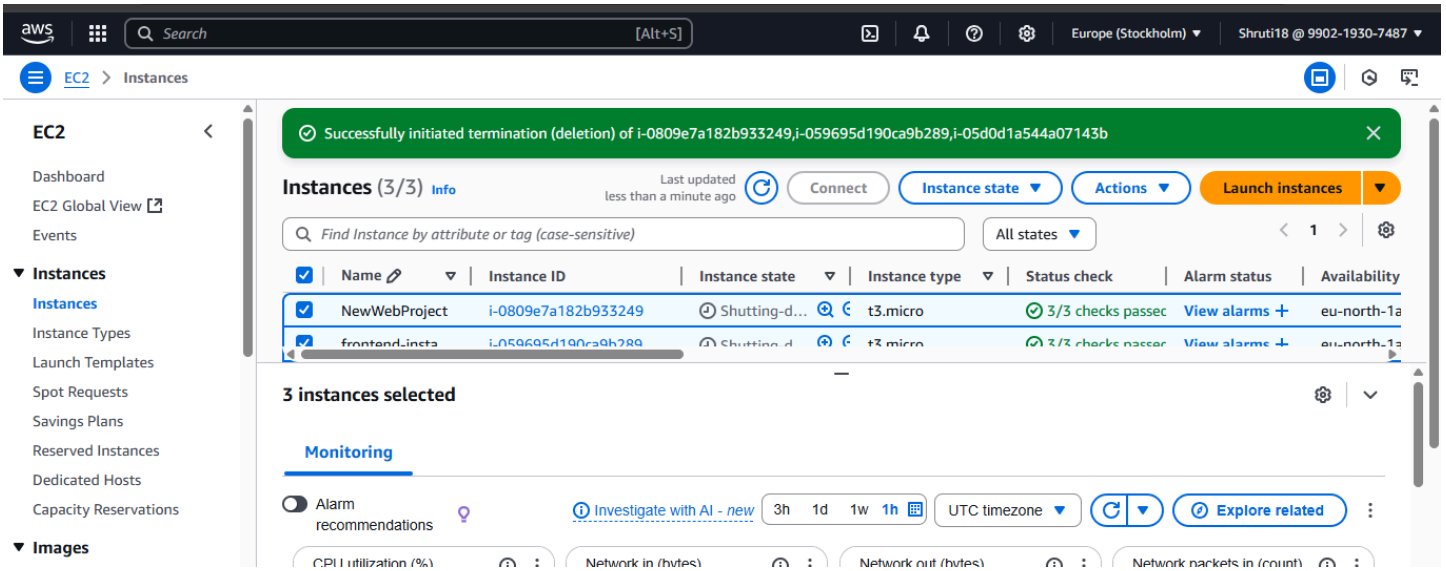
https://hub.docker.com/r/shruti1804/foodcritic-frontend
https://hub.docker.com/r/shruti1804/foodcritic-backend

```
ubuntu@ip-172-31-30-246:~/foodCriticApp$ sudo docker push shruti1804/foodcritic-frontend:latest
The push refers to repository [docker.io/shruti1804/foodcritic-frontend]
e56b311e5496: Pushed
8e3040133fd3: Pushed
0183b10b12bd: Pushed
b4fc2616e4e2: Pushed
cf7d2d64b3b8: Pushed
d2a991bcab4d: Mounted from library/node
b624aa2d5ea2: Mounted from library/node
d399c9dc306f: Mounted from library/node
84f9fa179c1b: Mounted from library/node
ce84ba212e49: Mounted from library/node
e4dc8cd9ecc8: Mounted from library/node
6428cc293366: Mounted from library/node
2f7436e79a0b: Mounted from library/node
latest: digest: sha256:c95554277909c890d4bf94fce73165a48267dc4909914a0e100cea231c658b0d size: 3055
ubuntu@ip-172-31-30-246:~/foodCriticApp$
```





Terminating the Instances:



Conclusion:

This project helped me understand how DevOps tools can be used together to build and run a complete application in the cloud. I learned how to use Terraform to create infrastructure, Docker to containerize applications, and Jenkins to automate the whole process step-by-step.

By dividing the project into three stages:

Creating infrastructure

Deploying the app

Testing.

I was able to manage everything in a clear and organized way. The frontend and backend worked properly on two different EC2 instances, and I successfully tested that the data entered in the frontend was saved in the backend database.

Overall, this project gave me hands-on experience in automation, cloud deployment, and DevOps practices that are used in real-world software development.